RAYCASTED: A RAYCAST END-TO-END DETECTION MODEL FOR LIGHTWEIGHT AND BOUNDARY-AWARE NUCLEI DETECTION IN HISTOPATHOLOGY IMAGES

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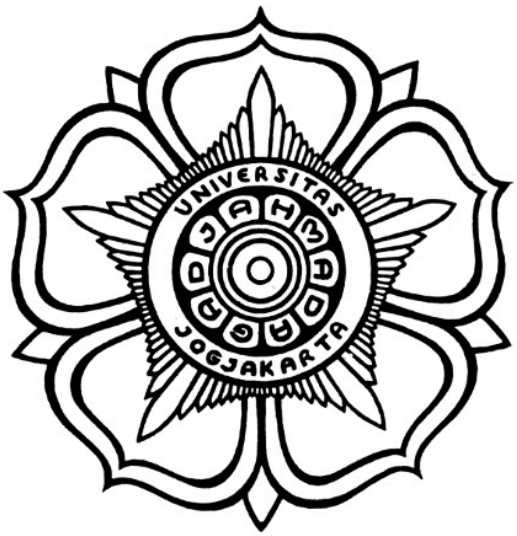

THE SUSTAINABLE DEVELOPMENT GOALS
Industry, Innovation and Infrastructure
Affordable and Clean Energy
Climate Action

Written by:

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BIOMEDICAL ENGINEERING PROGRAM

DEPARTMENT OF ELECTRICAL ENGINEERING AND INFORMATION TECHNOLOGY
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2026

ENDORSEMENT PAGE

RAYCASTED: A RAYCAST END-TO-END DETECTION MODEL FOR LIGHTWEIGHT AND BOUNDARY-AWARE NUCLEI DETECTION IN HISTOPATHOLOGY IMAGES

THESIS

Proposed as A Requirement to Obtain
Undergraduate Degree (*Sarjana Teknik*)
in Department of Electrical Engineering and Information Technology
Faculty of Engineering
Universitas Gadjah Mada

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Tuliskan kepada siapa skripsi ini dipersembahkan!

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PREFACE

Contoh: Puji syukur ke hadirat Allah SWT atas limpahan rahmat, karunia, serta petunjuk-Nya sehingga tugas akhir berupa penyusunan skripsi ini telah terselesaikan dengan baik. Dalam hal penyusunan tugas akhir ini penulis telah banyak mendapatkan arahan, bantuan, serta dukungan dari berbagai pihak. Oleh karena itu pada kesempatan ini penulis mengucapkan terima kasih kepada:

1. Orang 1 yang telah
2. Orang 2 yang telah
3. <isi dengan nama orang lainnya>

Akhir kata penulis berharap semoga skripsi ini dapat memberikan manfaat bagi kita semua, aamiin.

Catatan: setiap nama yang dituliskan boleh disertai dengan alasan berterima kasih.

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NOMENCLATURE AND ABBREVIATION

[SAMPLE]

b	=	bias
$K(x_i, x_j)$	=	fungsi kernel
y	=	kelas keluaran
C	=	parameter untuk mengendalaikan besarnya pertukaran antara penalti variabel slack dengan ukuran margin
L_D	=	persamaan Lagrange dual
L_P	=	persamaan Lagrange primal
w	=	vektor bobot
x	=	vektor masukan
ANFIS	=	Adaptive Network Fuzzy Inference System
ANSI	=	American National Standards Institute
DAG	=	Directed Acyclic Graph
DDAG	=	Decision Directed Acyclic Graph
HIS	=	Hue Saturation Intensity
QP	=	Quadratic Programming
RBF	=	Radial Basis Function
RGB	=	Red Green Blue
SV	=	Support Vector
SVM	=	Support Vector Machines

INTISARI

Intisari ditulis menggunakan bahasa Indonesia dengan jarak antar baris 1 spasi dan maksimal 1 halaman. Intisari sekurang-kurangnya berisi tentang latar belakang dan tujuan penelitian, metodologi yang digunakan, hasil penelitian, kesimpulan dan implikasi, dan Kata kunci yang berhubungan dengan penelitian.

Kata Kunci ditulis maksimal 5 kata yang paling berhubungan dengan isi skripsi. Silakan mengacu pada ACM / IEEE *Computing classification* jika Anda adalah mahasiswa Sarjana TI http://www.acm.org/about/class/ atau mengacu kepada IEEE keywords http://www.ieee.org/documents/taxonomy_v101.pdf jika Anda berasal dari Prodi Sarjana TE.

Kata kunci : Kata kunci 1, Kata kunci 2, Kata kunci 3, Kata kunci 4, Kata kunci 5

Contoh Abstrak Teknik Elektro:

"Penelitian ini bertujuan untuk mengembangkan sistem pengendalian suhu ruangan dengan menggunakan microcontroller. Metodologi yang digunakan adalah desain sirkuit, implementasi sistem pengendalian, dan pengujian performa. Hasil penelitian menunjukkan bahwa sistem pengendalian suhu ruangan yang dikembangkan mampu mengendalikan suhu ruangan dengan akurasi sebesar ±0,5°C. Kesimpulan dari penelitian ini adalah sistem pengendalian suhu ruangan yang dikembangkan efektif dan efisien.

Kata kunci: microcontroller, sistem pengendalian suhu, akurasi."

Contoh Abstrak Teknik Biomedis:

"Penelitian ini bertujuan untuk mengevaluasi keefektifan prototipe alat pemantau denyut jantung berbasis elektrokardiogram (ECG) untuk pasien jantung. Metodologi yang digunakan meliputi desain dan pembuatan prototipe, pengujian dengan pasien, dan analisis data. Hasil penelitian menunjukkan bahwa prototipe alat pemantau denyut jantung berbasis ECG memiliki akurasi yang baik dan mampu memantau denyut jantung pasien secara efektif. Kesimpulan dari penelitian ini adalah prototipe alat pemantau denyut jantung berbasis ECG merupakan solusi yang efektif dan efisien untuk memantau pasien jantung.

Kata kunci: elektrokardiogram, alat pemantau denyut jantung, akurasi."

Contoh Abstrak Teknologi Informasi:

"Penelitian ini bertujuan untuk mengevaluasi keamanan dan privasi pengguna aplikasi media sosial terpopuler. Metodologi yang digunakan meliputi analisis kebijakan privasi dan pengaturan keamanan, pengujian penetrasi, dan survei pengguna. Hasil penelitian menunjukkan bahwa beberapa aplikasi media sosial memiliki kebijakan privasi yang kurang jelas dan rendahnya tingkat keamanan. Kesimpulan dari penelitian ini adalah pentingnya meningkatkan kebijakan privasi dan tingkat keamanan pada aplikasi media sosial untuk melindungi privasi dan data pengguna.

Kata kunci: media sosial, keamanan, privasi, pengguna."

ABSTRACT

Abstract ditulis italic (miring) menggunakan bahasa Inggris dengan jarak antar baris 1 spasi dan maksimal 1 halaman. Abstract adalah versi Bahasa Inggris dari intisari. Abstract dapat ditulis dalam beberapa paragraf. Baris pertama paragraph harus menjorok ke dalam sekitar 1 cm. Tidak dsarankan menggunakan mesin penerjemah melainkan tulis ulang.

Keywords : Keyword 1, Keyword 2, Keyword 3, Keyword 4, Keyword 5

CHAPTER I
INTRODUCTION

1.1 Background

Nuclei segmentation is a critical first step in the analysis of histopathological images from Whole Slide Images (WSIs), as it underpins many downstream processes such as tumor grading, cellular composition analysis, and biomarker quantification [1]. In recent years, deep learning (DL) has become the staple approach for performing nuclei segmentation, driven by its ability to learn hierarchical features directly from data [1, 2]. Despite these advances, one of the principal difficulties that DL models continue to face is the presence of clumped nuclei, which results in overlapping and touching instances that are difficult to separate accurately.

Instance segmentation has emerged as the dominant paradigm for addressing this challenge, owing to its ability to classify individual objects as unique instances [3]. Among instance segmentation approaches, mask-based methods are the preferred choice in the literature [4]. However, mask-based instance segmentation carries notable drawbacks. First, these methods are computationally heavy, typically relying on two-stage architectures or elaborate post-processing pipelines [5]. Second, mask-based representations enforce pixel exclusivity: each pixel can belong to only one segment, which distorts morphology at contact boundaries where cells overlap [6].

An alternative line of work is contour-based instance segmentation, which represents objects through their boundary geometry rather than dense pixel masks. Contour-based methods offer two key advantages: they are inherently lighter, often employing single-stage architectures [5], and they can incorporate geometric priors into the representation prior to training [7]. Nevertheless, existing contour-based approaches remain dependent on non-maximum suppression (NMS) as a post-processing step, which adds computational overhead and risks removing valid predictions when instances are in close proximity [8]. Although vision transformer (ViT) based methods can eliminate the NMS dependency, they introduce substantially higher computational cost that limits their practicality.

This research proposes, **RayCastED**, a deep learning model that bridges the gap between these paradigms by combining three desirable properties. *Geometry-awareness:* the model employs ray-prediction based on star-convex polygons to capture true cellular boundaries, enabling precise area and morphological measurements without classifying every pixel. *Lightweight architecture:* the FCN backbone keeps computational overhead low, making the model suitable for edge deployment via TensorRT and ONNX Runtime. *Truly end-to-end:* the architecture completely eliminates the need for NMS or other

post-processing, allowing it to natively and accurately resolve dense, highly overlapping cellular scenarios while reducing inference overhead.

1.2 Problem Statement

Based on the background discussed above, the following problems are identified:

1. Touching or overlapping nuclei introduce computational overhead.
   Existing methods address clumped nuclei either through two-stage architectures, heavy post-processing pipelines such as NMS, or computationally expensive vision transformer (ViT) backbones. Each of these strategies imposes a significant computational burden that limits scalability to large WSIs.
2. State-of-the-art methods are not suitable for edge deployment
   Current leading approaches to nuclei segmentation and detection are designed with computational assumptions that exceed the capabilities of resource-constrained edge devices. Deploying these models in clinical settings requires hardware that is often unavailable at the point of care.

1.3 Objectives

To address the identified problems, this thesis has the following objectives:

1. Develop **RayCastED** and evaluate the performance by comparing it against prior nuclei instance segmentation models.
2. Evaluate the deployment feasibility of RayCastED on edge devices.

1.4 Research Scope

This research is bounded by the following scope limitations:

1. **Data characteristics:** The study uses only H&E stained histopathology images with approximately 0.25 MPP (microns per pixel) resolution from publicly available secondary datasets. The datasets encompass multiple tissue types/organs.
2. **Computational constraints:** Model training is performed on RTX 5070Ti hardware. Baseline comparisons are limited to models compatible with this hardware configuration.
3. **Validation approach:** The research employs computational validation using standard metrics. Pathologist or expert validation is not included in this study.
4. **Technical boundaries:** There are no limitations on nuclei types, sizes, or detection scope. No strict constraints are imposed on input patch sizes or memory usage during model training and inference.

1.5 Research Contributions

The contributions of this research are as follows:

1. **RayCastED model implementation:** A complete implementation of the proposed star-convex polygon-based detection model with raycast representation and end-to-end training, built on a fully convolutional backbone.
2. **Performance benchmarks on standard datasets:** Comprehensive evaluation of RayCastED on publicly available histopathology datasets, comparing performance against baseline models including mask-based instance segmentation, contour-based methods, and transformer-based approaches.
3. **Edge deployment validation:** Validation of RayCastED's deployment capabilities on edge devices through TensorRT and ONNX Runtime conversion, demonstrating practical feasibility for resource-constrained clinical environments.
4. **Systematic comparison against baselines:** Quantitative comparison with prior nuclei detection and segmentation methods to measure improvements in boundary accuracy, counting precision in dense regions, and computational efficiency.
5. **Reproducible codebase:** A publicly available, well-documented implementation that enables reproducibility and provides a foundation for future research in polygon-based detection for computational pathology.

1.6 Presentation Structure

This thesis is organized into five chapters. Chapter 1 introduces the background, problem statement, objectives, and scope of the research. Chapter 2 reviews the literature on nuclei instance segmentation paradigms, the theoretical foundations of the proposed model components, and the evaluation metrics used throughout this work. Chapter 3 describes the methodology, covering datasets, the RayCast representation, the data preprocessing pipeline, the proposed model architecture components, and the training strategy. Chapter 4 presents the experimental results, including the RayCast representation validation, benchmarking against prior methods, edge deployment evaluation, and ablation studies. Chapter 5 concludes the thesis with a summary of findings and directions for future work.

CHAPTER II

LITERATURE REVIEW AND THEORETICAL BASIS

2.1 Literature Review

2.1.1 State-of-the-Art Literature Review

Table 2.1 summarizes the key prior works most relevant to the proposed RayCastED model.

Table 2.1. Literature review of prior nuclei instance segmentation and detection methods.

Method	Year	Author	Approach	Mechanism
StarDist [8, 9]	2018	Schmidt et al.	contour-based	Star-convex polygon; predicts radial distances from centroid to boundary at fixed angular intervals; NMS post-processing
SplineDist [10]	2020	Mandal & Uhlmann	contour-based	B-spline control points for cell boundary contours; infers cell shape from radial distances and fits splines; requires centroid-anchored patches; watershed or connected-components post-processing
TSFD-Net [11]	2022	Ilyas et al.	mask-based	Tissue-specific feature distillation backbone extracts morphological features varying by tissue type; BiFPN generates hierarchical feature pyramid; interlinked decoders jointly optimize and fuse features; combinational loss for joint nuclei segmentation and classification; NMS post-processing
CPP-Net [12]	2023	Chen et al.	contour-based	Context-aware polygon proposal network; predicts polygon vertices directly with multi-scale feature aggregation; NMS post-processing

Continued on next page

Table 2.1 – continued from previous page

Method	Year	Author	Approach	Mechanism
CellViT [13]	2023	Hörst et al.	mask-based	Vision Transformer encoder pre-trained on 104M histological image patches; leverages Segment Anything Model foundational architecture; transformer-based segmentation decoder; predicts cell probability, distance maps, and class labels; watershed post-processing
HoVer-NeXt [14, 15]	2024	Baumann et al.	mask-based	Horizontal/vertical distance maps from each pixel to neighboring centroids; watershed + morphological post-processing for instance separation
LKCell [16]	2024	Cui et al.	mask-based	Large convolution kernels for computationally efficient receptive fields; transfers pre-trained large-kernel CNN to medical domain; single decoder with NP/HV/NT multi-task output heads; watershed post-processing
CellPose [17, 18]	2020	Stringer et al.	mask-based	Gradient flow field pointing toward cell centroids combined with cell probability map; flow-field post-processing

2.1.2 Research Gap and Novelty

The novelty of this research lies in the development of RayCastED, which proposes a deep learning architecture combining three elements: (1) a star-convex polygon representation that captures cell boundary geometry without dense pixel-wise classification, (2) a fully convolutional network backbone that maintains a lightweight computational profile suitable for edge deployment, and (3) a truly end-to-end design that eliminates NMS dependency entirely.

As summarized in Table 2.1, all eight prior methods share two key limitations: they rely on dense per-pixel prediction and require handcrafted post-processing (NMS, watershed, or flow-field) to produce final instance masks. RayCastED addresses this gap

by providing end-to-end, NMS-free nuclei detection with a lightweight fully convolutional architecture, bridging the morphological accuracy of contour-based segmentation with the computational efficiency of single-shot detection for nuclei analysis in histopathology images.

2.2 Theoretical Basis

2.2.1 Histopathological Foundations

Histopathology is the study of tissue disease through microscopic examination of tissue samples. In clinical practice, tissue specimens are collected through biopsy or surgical resection and processed for microscopic analysis. The tissue undergoes fixation (typically in formalin) to preserve cellular structures, followed by embedding in paraffin wax, sectioning into thin slices, and mounting on glass slides for staining and visualization.

The cell nucleus is a membrane-bound organelle containing genetic material and serves as the primary structure of interest in histopathological analysis. Nuclear morphology, including size, shape, texture, and spatial distribution, provides critical diagnostic information. In cancer diagnosis, malignant cells often exhibit nuclear atypia: enlarged nuclei, irregular shapes, hyperchromasia (increased staining intensity), and abnormal mitotic figures.

2.2.1.1 Nuclear Morphology and Shape Characteristics

The shape of cell nuclei varies considerably across tissue types and pathological states, but the majority of nuclei encountered in routine H&E-stained histopathology sections exhibit a defining geometric property: they are **approximately convex**. Healthy epithelial nuclei are typically round or slightly oval, with smooth contours and minimal indentations. Lymphocytes are among the most regular, appearing as compact circles with diameters of approximately 5–8 μm. Fibroblast and endothelial nuclei tend to be elongated (spindle-shaped), but remain convex with no concavities. Even in malignant tissue, where nuclear pleomorphism introduces greater variability in size and eccentricity, most nuclei retain a fundamentally convex silhouette. Concavities do arise in practice, most commonly at the boundary between two overlapping nuclei where the cells appear to "press into" one another, but these are artifacts of the 2D projection rather than intrinsic features of the nuclear membrane itself.

This near-ubiquitous convexity has an important implication for automated representation. Because concave shapes are rarely encountered in their undistorted form, nuclear boundaries can be fully described by a centroid and a set of radial distances measured from that centroid to the boundary at regular angular intervals. This radial representation captures the complete boundary geometry with a compact set of parameters, avoiding both the redundancy of dense per-pixel segmentation and the information loss of

axis-aligned bounding boxes. The few cases where concavities do appear, primarily at overlap boundaries, are artifacts of the 2D projection and can be addressed by representing each overlapping nucleus independently rather than forcing a single non-overlapping mask.

2.2.1.2 The Overlapping Nuclei Problem

A fundamental challenge in nuclei analysis is that nuclei in tissue sections are densely packed and frequently overlap, forming clumped clusters where individual boundaries are difficult to distinguish even for expert pathologists. This overlapping occurs because tissue sections are typically cut at a thickness of 3–5 μm, which is comparable to or smaller than the diameter of many nuclei (typically 5–10 μm). As a result, multiple nuclei at different depths within the section are projected onto the same 2D image plane, creating the appearance of clumped or touching nuclei. This is not merely an artifact of imaging resolution but an inherent property of projecting 3D tissue architecture onto a 2D plane, motivating the development of automated methods capable of both accurately delineating individual nuclear boundaries and correctly separating touching or overlapping nuclei.

2.2.2 Digital Pathology Imaging

2.2.2.1 Tissue Slicing and Scanning

The digitization of histopathology slides produces whole-slide images (WSIs) through specialized scanners that capture high-resolution images of the entire tissue section. Modern slide scanners, such as those from Hamamatsu, Leica, and 3DHistech, use line-sensor or area-sensor cameras mounted on motorized stages to systematically capture overlapping tiles that are subsequently stitched into a single large image.

WSIs are typically stored in multi-resolution pyramidal formats (e.g., Aperio SVS, Hamamatsu NDP, or generic TIFF), where each level provides a different magnification. The spatial resolution of a WSI is specified in microns per pixel (MPP), which depends on the objective lens magnification used during scanning. Common resolutions range from approximately 0.25 MPP (equivalent to 40$\times$ magnification) to 0.5 MPP (20$\times$). At 0.25 MPP, a typical biopsy slide can produce an image exceeding 50,000 $\times$ 50,000 pixels, necessitating patch-based processing strategies for computational analysis.

2.2.2.2 Histopathological Slides Staining

Tissue staining is essential because most biological tissues are nearly transparent under standard brightfield microscopy. Stains introduce contrast by selectively binding to tissue components through chemical interactions. The most widely used staining protocol in histopathology is Hematoxylin and Eosin (H&E) staining.

Hematoxylin is a basic (positively charged) dye that binds to acidic structures such as DNA and RNA, coloring nuclei dark blue or purple. Eosin is an acidic (negatively charged) dye that binds to basic structures such as cytoplasmic proteins and extracellular matrix, coloring them pink. The combination produces the characteristic appearance of H&E-stained tissue: blue-purple nuclei against a pink cytoplasmic and stromal background.

Beyond H&E, several specialized stains are used in clinical practice: Periodic Acid-Schiff (PAS) for basement membranes and glycogen, Masson's Trichrome for collagen fibers, and immunohistochemical (IHC) stains for specific protein markers. Each stain produces distinct color characteristics that can be mathematically decomposed for computational analysis.

2.2.2.3 Beer-Lambert Law and Optical Density

The interaction of light with stained tissue follows the Beer-Lambert law, which describes the relationship between light absorption and the concentration of absorbing material. For a stained tissue sample, the transmitted intensity I at a given wavelength is related to the incident intensity I_0 by:

$$I = I_0 \cdot e^{-\alpha \cdot c \cdot d} \tag{2-1}$$

where α is the molar absorptivity (extinction coefficient), c is the concentration of the stain, and d is the path length through the tissue. The optical density (OD) is defined as:

$$\text{OD} = -\log_{10}\left(\frac{I}{I_0}\right) = \alpha \cdot c \cdot d \cdot \log_{10}(e) \tag{2-2}$$

In the context of digital pathology, the RGB pixel values of a scanned image are proportional to transmitted light intensity. Converting from RGB to optical density space is essential because OD is linearly additive: if multiple stains are present, the total OD at each pixel equals the sum of the OD contributions from each individual stain. This linearity property underpins color deconvolution methods.

2.2.2.4 Stain Vectors

A stain vector represents the characteristic color signature of a specific stain in optical density space. For an RGB image, each stain is described by a 3-element unit vector $\mathbf{s} = [s_R, s_G, s_B]^T$ in OD space, where each component indicates the stain's relative absorption at the red, green, and blue channels.

For H&E staining, the two stain vectors are:

- **Hematoxylin vector** $\mathbf{s}_H$: characterizes the absorption pattern of hematoxylin, which strongly absorbs in the blue-green range, producing its characteristic dark blue appearance.
- **Eosin vector** $\mathbf{s}_E$: characterizes the absorption pattern of eosin, which absorbs primarily in the green range, producing its pink appearance.

When two stains are present, the stain matrix S is formed by concatenating the stain vectors as columns: $S = [\mathbf{s}_H \quad \mathbf{s}_E]$. For three-stain protocols, a third column is added.

2.2.2.5 Stain Vector Estimation Methods

Since stain appearance varies across scanners, laboratories, and tissue types, the stain vectors must be estimated from each individual image or dataset. Several methods exist for automatic stain vector estimation:

Macenko's method [19] estimates stain vectors by analyzing the angular distribution of pixel OD values through Singular Value Decomposition (SVD). The procedure operates as follows.

Given an RGB image with N pixels, each pixel $\mathbf{p}_i = [R_i, G_i, B_i]^T$ is first converted to optical density space:

$$\mathbf{o}_i = -\log_{10}\left(\frac{\mathbf{p}_i}{255} + \epsilon\right) \tag{2-3}$$

where ϵ is a small constant to avoid logarithm of zero. Pixels whose OD magnitude $\|\mathbf{o}_i\|$ falls below a threshold t_{OD} are discarded, as they correspond to near-white background regions with negligible stain absorption. The threshold is typically set at the 1st percentile of the OD magnitude distribution.

For the remaining M pixels, a matrix $\hat{O} \in \mathrm{R}^{3\times M}$ is formed by arranging the OD vectors as columns. SVD is then applied:

$$\hat{O} = U\Sigma V^T \tag{2-4}$$

where $U \in \mathrm{R}^{3\times 3}$ contains the left singular vectors. The two principal singular vectors $\mathbf{u}_1$ and $\mathbf{u}_2$ (the first two columns of U) span the plane in which the stain vectors lie. Each OD vector is projected onto this plane:

$$\mathbf{p}'_i = \begin{bmatrix} \mathbf{u}_1^T \mathbf{o}_i \\ \mathbf{u}_2^T \mathbf{o}_i \end{bmatrix} \tag{2-5}$$

and converted to angular coordinates $\theta_i = \arctan 2(p'_{i,2},\ p'_{i,1})$. The stain directions are determined by finding the angles $\theta_{\min}$ and $\theta_{\max}$ that enclose the 1st and 99th percentiles of

the angular distribution, corresponding to the extreme concentrations of each stain.

The raw stain vectors are then reconstructed in OD space:

$$\mathbf{v}_1 = \mathbf{u}_1 \cos\theta_{\min} + \mathbf{u}_2 \sin\theta_{\min}, \quad \mathbf{v}_2 = \mathbf{u}_1 \cos\theta_{\max} + \mathbf{u}_2 \sin\theta_{\max} \tag{2-6}$$

Since SVD produces vectors with arbitrary sign, the vectors are flipped such that all components are non-negative (as OD values are inherently non-negative). Finally, the vectors are normalized to unit length: $\mathbf{s}_j = \mathbf{v}_j / \|\mathbf{v}_j\|$.

To assign identities, the angle each normalized vector makes with a predefined reference direction in OD space (typically corresponding to a hematoxylin-like hue) is computed. The vector closer to the hematoxylin reference is labeled $\mathbf{s}_H$, and the other is labeled $\mathbf{s}_E$.

A residual stain vector is then computed via the cross product of the two estimated stain vectors:

$$\mathbf{s}_R = \mathbf{s}_H \times \mathbf{s}_E \tag{2-7}$$

This third vector is orthogonal to both $\mathbf{s}_H$ and $\mathbf{s}_E$ and captures the residual OD component not explained by the two primary stains, arising from background variation, optical noise, or imaging artifacts. It is included as the third column of the stain matrix to form a complete basis $S = [\mathbf{s}_H \ \mathbf{s}_E \ \mathbf{s}_R] \in \mathrm{R}^{3\times3}$, which is required for the matrix inversion in color deconvolution (Section2.2.2.6). Without this third vector, the stain matrix would be rank-deficient and non-invertible.

Macenko's method is widely adopted due to its computational simplicity and robustness, as the SVD-based projection efficiently captures the dominant stain directions without iterative optimization.

2.2.2.6 Color Deconvolution

Color deconvolution is the process of separating an H&E-stained image into its individual stain channels. Given the stain matrix S and the OD image $\mathbf{v}$ (a pixel vector in OD space), the stain concentrations $\mathbf{c}$ for each pixel are obtained by solving the linear system:

$$\mathbf{v} = S \cdot \mathbf{c} \tag{2-8}$$

Since S is a 3×2 matrix (for H&E) or 3×3 (for three stains), the system can be solved using the pseudoinverse:

$$\mathbf{c} = S^{-1} \cdot \mathbf{v} \tag{2-9}$$

where S^{-1} denotes the pseudoinverse (or inverse, for square S). The resulting stain concentration maps $\mathbf{c}$ provide separate grayscale images representing the hematoxylin and eosin density at each pixel. The hematoxylin channel is particularly useful for nuclei detection algorithms, as it isolates nuclear staining from cytoplasmic and stromal background.

2.2.2.7 Stain Normalization

Stain normalization addresses the significant color variation that exists across WSIs captured from different scanners, hospitals, or staining batches. This variation, if uncorrected, degrades the performance of deep learning models trained on data from one source and applied to another.

The standard approach to stain normalization involves three steps: (1) estimate the stain matrix S_{source} from the source image using one of the methods described above, (2) compute the stain concentration map $\mathbf{c} = S_{\text{source}}^{-1} \cdot \mathbf{v}_{\text{source}}$, and (3) reconstruct the normalized image using a target stain matrix S_{target}: $\mathbf{v}_{\text{normalized}} = S_{\text{target}} \cdot \mathbf{c}$, then convert back from OD to RGB space.

This process preserves the structural content of the image while standardizing color appearance. Stain normalization is commonly applied as a preprocessing step before training or inference in computational pathology pipelines.

2.2.3 Instance Segmentation Paradigms

Instance segmentation extends semantic segmentation by assigning a unique label to each individual object instance, producing both a class label and an instance identity for every pixel. This section surveys the two dominant paradigms for nuclei instance segmentation: mask-based methods that produce dense per-pixel predictions, and contour-based methods that predict sparse geometric parameters.

2.2.3.1 Mask-Based Instance Segmentation

Two dominant paradigms exist for mask-based instance segmentation. **Proposal-based methods** such as Mask R-CNN [20] operate in two stages: a region proposal network (RPN) generates candidate object bounding boxes, and a mask head then predicts a binary segmentation mask for each proposal. While accurate, the two-stage pipeline introduces significant computational overhead, particularly when the number of instances is large.

Dense prediction methods such as U-Net [21] and HoVer-Net [15] predict instance information directly from dense feature maps. U-Net produces pixel-level embeddings where each pixel is mapped to a high-dimensional space, and instances are separated

by clustering pixels based on embedding distance. HoVer-Net predicts horizontal and vertical distance maps from each pixel to the centroids of neighboring cells, using these maps to separate touching or overlapping instances. While these methods can achieve high accuracy, they require computationally expensive post-processing (e.g., watershed algorithms, clustering, or morphological operations) to convert dense predictions into individual instance masks.

Mask-based approaches face two fundamental limitations when applied to nuclei analysis in histopathology images.

The **overlap problem** arises from the mutually exclusive pixel assignment inherent in segmentation. In a two-dimensional image, each pixel can only belong to one instance. When nuclei overlap in 3D tissue and are projected onto a 2D plane, their boundaries become entangled: a single pixel at the boundary between two overlapping nuclei cannot simultaneously represent both cells. Segmentation models resolve this ambiguity by drawing an artificial boundary between the two cells, which distorts the true morphology of both. This distortion is particularly problematic for nuclei analysis, where accurate boundary delineation directly affects area-based measurements such as TILs scoring.

The **computational bottleneck** stems from the dense prediction paradigm: segmentation models must classify every pixel in the image, including the vast majority of background pixels that carry no useful information. For a typical histopathology patch of 256×256 pixels, this means producing 65,536 individual predictions, even when only a small fraction correspond to actual nuclei. This per-pixel cost becomes prohibitive when processing whole-slide images, which may contain billions of pixels. The post-processing required to extract individual instances from dense predictions (clustering, watershed, morphological operations) adds further computational overhead.

2.2.3.2 Contour-Based Instance Segmentation

Contour-based instance segmentation approaches the problem from a different perspective. Instead of predicting a dense pixel-wise mask, contour-based methods predict a sparse set of geometric parameters that describe each object's boundary. The most prominent representation for nuclei is the star-convex polygon [8], where a nucleus is represented by its centroid and a set of radial distances at fixed angular intervals. This representation is compact (requiring only $2 + R$ parameters per nucleus, where R is the number of rays), inherently boundary-aware, and can naturally handle overlapping instances by allowing polygons to intersect.

Contour-based instance segmentation shares its architectural structure with object detection rather than with mask-based segmentation. Both contour-based methods and detection operate on an anchor grid, predict a sparse set of geometric parameters per location, and train through matching mechanisms rather than per-pixel classification. The

detection foundations that enable contour-based methods are discussed below.

2.2.3.2.1 Anchor Grid

In grid-based detection frameworks, predictions are organized around an **anchor grid**: a regular lattice of reference boxes (anchors) placed at each spatial location of a feature map. At each grid cell, multiple anchors of different scales and aspect ratios are defined to accommodate objects of varying sizes and shapes. Each anchor serves as a prior that the network refines into a final prediction by regressing coordinate offsets relative to the anchor dimensions and classifying the object category.

The anchor grid determines several key properties of the detector. It defines the maximum number of detections the model can produce: for a feature map of resolution $H \times W$ with A anchors per cell, the model generates up to $H \times W \times A$ predictions. Each grid cell is responsible for detecting objects whose centroids fall within its receptive field, creating a natural spatial correspondence between image regions and model outputs. This regular grid structure enables the detector to process objects at multiple scales simultaneously by operating on feature maps at different resolutions, with coarser grids detecting larger objects and finer grids detecting smaller objects.

2.2.3.2.2 Matching Mechanisms in Object Detection

A fundamental challenge in training detection models is establishing the correspondence between predicted outputs and ground-truth objects, a process known as **matching**. Given the anchor grid produces a large number of predictions per image, the matching mechanism determines which predictions are assigned to which ground-truth objects during loss computation.

During training, predictions are matched to ground-truth objects using an assignment rule. The assignment is formulated by constructing a **cost matrix** (also called a score matrix, depending on whether lower or higher values indicate better matches) that quantifies the quality of each prediction–ground-truth pair. Given N ground-truth objects and M predictions, the cost matrix $\mathbf{C} \in \mathbb{R}^{N \times M}$ is defined such that element $C_{i,j}$ measures the dissimilarity between ground-truth object i and prediction j. The most common choice for $C_{i,j}$ is the negative IoU:

$$C_{i,j} = 1 - \text{IoU}(g_i, p_j) \tag{2-10}$$

where g_i is the bounding box (or polygon) of ground-truth object i and p_j is the predicted bounding box (or polygon). Since IoU ranges from 0 to 1, the cost $C_{i,j}$ also ranges from 0 (perfect overlap) to 1 (no overlap). Alternatively, some frameworks use a score matrix $\mathbf{S} \in \mathbb{R}^{N \times M}$ where $S_{i,j} = \text{IoU}(g_i, p_j)$ and the goal is to maximize the total

score rather than minimize the total cost. The two formulations are equivalent: minimizing $\sum C_{i,j}$ is the same as maximizing $\sum S_{i,j}$ when $C_{i,j} = 1 - S_{i,j}$.

In addition to localization cost, the cost matrix may incorporate a **classification cost** term that penalizes mismatched class predictions. For multi-class detection with K classes, the classification cost can be defined using the cross-entropy between the predicted class distribution and the ground-truth class:

$$C_{i,j}^{\text{cls}} = -\log \hat{p}_j(c_i) \tag{2-11}$$

where $\hat{p}_j(c_i)$ is the predicted probability of class c_i (the true class of ground-truth i) and the total cost combines both terms:

$$C_{i,j} = C_{i,j}^{\text{loc}} + \lambda_{\text{cls}} \cdot C_{i,j}^{\text{cls}} \tag{2-12}$$

with λ_{cls} controlling the relative weight of classification versus localization. This combined cost matrix encodes both spatial alignment and semantic correctness into a single assignment objective.

Once the cost matrix is constructed, an assignment algorithm operates on it to determine which predictions correspond to which ground-truth objects. The choice of assignment algorithm determines whether the matching is one-to-many (allowing multiple predictions per ground-truth object), many-to-one (allowing one prediction to be matched to multiple ground-truth objects), or one-to-one (bipartite). Traditional methods such as Faster R-CNN [22] use IoU-based thresholding on this matrix: a prediction is assigned to a ground-truth object if $C_{i,j}$ falls below a threshold. This assignment is often ambiguous when multiple predictions overlap with the same ground-truth object, requiring heuristics to resolve conflicts. Modern YOLO variants use more sophisticated strategies, such as task-aligned matching and dual-label assignment, which extend the cost matrix with additional alignment terms. These strategies are discussed in Section 2.2.8.

2.2.3.2.3 Feature Pyramid Networks

A single feature map at a fixed resolution presents a fundamental limitation for object detection: objects in natural images vary widely in scale, and features that are discriminative for large objects (derived from deep, low-resolution layers with large receptive fields) may be too coarse to localize small objects, while features suitable for small objects (from shallow, high-resolution layers) may lack the semantic context needed to recognize large objects. **Feature Pyramid Networks (FPN)** [23] address this by constructing a multi-scale feature pyramid from a single input image in a computationally efficient manner.

An FPN extends a standard convolutional backbone with a top-down pathway and lateral connections. The backbone computes a hierarchy of feature maps $\{C_2, C_3, C_4, C_5\}$ at progressively coarser resolutions (downsampled by factors of 4, 8, 16, and 32 relative to the input). The top-down pathway then upsamples the deepest feature map C_5 and merges it with the corresponding lateral feature map from the backbone (e.g., C_4) via element-wise addition after applying a 1×1 convolution to match channel dimensions. This process is repeated to produce the pyramid levels $\{P_2, P_3, P_4, P_5\}$:

$$P_\ell = \text{Conv}_{1\times1}(C_\ell) + \text{Upsample}(P_{\ell+1}) \tag{2-13}$$

Each pyramid level P_ℓ has the same spatial resolution as its corresponding backbone feature map C_ℓ but benefits from the semantic richness of higher-level features propagated downward through the top-down pathway.

In a detector, each pyramid level is assigned to objects within a specific scale range. Large objects are detected at coarser levels (P_5) where the large receptive field captures holistic shape, while small objects are detected at finer levels (P_2) where high spatial resolution preserves precise localization. This per-level assignment reduces ambiguity in the matching process: objects of clearly different scales are assigned to different pyramid levels, preventing them from competing for the same prediction slot at a single resolution.

However, FPN also contributes to duplicate predictions. When an object falls near the boundary between two assigned scale ranges, it may be detected at two adjacent pyramid levels simultaneously, producing two spatially overlapping predictions for the same object. These cross-level duplicates are a significant source of redundancy that downstream NMS must resolve.

2.2.3.2.4 Non-Maximum Suppression

Non-maximum suppression (NMS) is a post-processing step used by most detection frameworks to eliminate duplicate predictions. When multiple predictions overlap with the same ground-truth object, NMS selects the prediction with the highest confidence score and suppresses (removes) all other predictions whose overlap with the selected one exceeds a predefined IoU threshold.

The standard NMS procedure operates as follows. Given a set of detections ranked by confidence score, the highest-confidence detection is selected and all remaining detections with IoU exceeding a threshold τ_{nms} are suppressed. This process repeats until no detections remain.

Formally, let $\mathcal{D} = \{d_1, d_2, \ldots, d_N\}$ be a set of detections sorted in descending order of confidence s_i, each with bounding box B_i. The NMS-filtered output set $\mathcal{D}'$ is

defined inductively:

$$\mathcal{D}' = \{d_i \in \mathcal{D} : \forall\, d_j \in \mathcal{D}' \text{ with } s_j > s_i,\ \text{IoU}(B_i, B_j) < \tau_{\text{nms}}\} \tag{2-14}$$

Alternatively, expressed as a greedy iteration that produces a maximal independent set with respect to the τ_{nms}-thresholded neighborhood graph: starting from $\mathcal{D}' = \emptyset$, for each $d_i \in \mathcal{D}$ in confidence order, if $\max_{d_j \in \mathcal{D}'} \text{IoU}(B_i, B_j) < \tau_{\text{nms}}$, then $\mathcal{D}' \leftarrow \mathcal{D}' \cup \{d_i\}$.

The IoU threshold τ_{nms} controls the trade-off between eliminating duplicates and preserving valid detections of closely spaced objects.

NMS introduces a fundamental tension in nuclei detection. In densely clumped regions, multiple nuclei overlap significantly, producing predictions that are spatially close and have high mutual IoU. A low NMS threshold aggressively suppresses these predictions, reducing duplicates but also removing valid detections of distinct cells. A high threshold preserves more predictions but allows duplicates to remain, inflating the cell count. Neither setting resolves the underlying problem: NMS cannot distinguish between two overlapping predictions of the same cell and two valid predictions of adjacent cells that happen to overlap.

2.2.3.2.5 Limitations for Nuclei Analysis

Contour-based methods inherit the architectural properties of object detection, including its limitations. While the star-convex polygon representation addresses the boundary problem by preserving cell morphology, the **density problem** persists. In densely clumped regions, neighboring nuclei generate overlapping polygon predictions, and NMS must distinguish between duplicate predictions of the same cell and valid predictions of distinct cells. Since NMS relies solely on geometric overlap and confidence scores, it systematically undercounts in clumps where adjacent cells have high mutual IoU. This creates a density-dependent counting bias where the model performs worst precisely in the regions most relevant to clinical assessment.

2.2.4 Convolutional Neural Networks

2.2.4.1 Convolution and Feature Extraction

A convolutional layer applies a set of learnable filters (kernels) to an input feature map, producing an output feature map where each element is a weighted sum of local input elements. For a 2D convolution with input $\mathbf{X} \in \mathbb{R}^{H \times W \times C_{\text{in}}}$ and kernel $\mathbf{W} \in \mathbb{R}^{K \times K \times C_{\text{in}} \times C_{\text{out}}}$, the output at spatial position (i, j) for channel c is:

$$\mathbf{Y}_{i,j,c} = \sum_{m=0}^{K-1} \sum_{n=0}^{K-1} \sum_{k=0}^{C_{\text{in}}-1} \mathbf{W}_{m,n,k,c} \cdot \mathbf{X}_{i+m,j+n,k} + b_c \tag{2-15}$$

where b_c is the bias term for output channel c. Each kernel slide across the input computes the same operation at every spatial position, exploiting the translation invariance of visual patterns. Stacking multiple convolutional layers with nonlinear activation functions (typically ReLU) enables the network to learn hierarchical feature representations: early layers capture low-level features such as edges and textures, while deeper layers capture higher-level structures such as cell shapes and tissue patterns.

2.2.4.2 Receptive Field and Kernel Size

The receptive field of a neuron in a convolutional network is the region of the input image that influences its output. For a single convolutional layer with kernel size K, each neuron has a receptive field of $K \times K$ pixels. Stacking L convolutional layers, each with kernel size K and stride 1, produces a receptive field of size $(K + (K-1)(L-1)) \times (K + (K-1)(L-1))$ at the deepest layer. Using stride $s > 1$ or pooling layers between convolutions further enlarges the effective receptive field.

The kernel size directly determines the model's ability to capture spatial context. A 3×3 kernel sees only immediate neighbors, while a 7×7 kernel captures a larger local context. However, larger kernels increase the number of parameters quadratically: a 7×7 kernel has 5.4 times more parameters than a 3×3 kernel for the same number of channels. Modern architectures typically use small kernels (3×3) stacked deeply to achieve large effective receptive fields while keeping the parameter count manageable.

For nuclei detection, a sufficiently large receptive field is critical because nuclei can vary significantly in size, and contextual information from surrounding tissue helps distinguish nuclei from artifacts and debris. Methods such as LKCell [16] demonstrate that using large convolution kernels (e.g., 7×7 or larger) with structural reparameterization can achieve large receptive fields with lower computational cost than stacking many small-kernel layers.

2.2.4.3 Depthwise Convolution

Depthwise convolution is a spatial decomposition of the standard convolution operation that significantly reduces computational cost. In a standard convolution with C_{in} input channels and C_{out} output channels using a $K \times K$ kernel, the number of multiplications per spatial position is $K^2 \cdot C_{\text{in}} \cdot C_{\text{out}}$. Depthwise convolution decomposes this into two steps:

1. **Depthwise step:** Each of the C_{in} input channels is convolved independently with its own $K \times K$ kernel, producing C_{in} intermediate feature maps. Cost: $K^2 \cdot C_{\text{in}}$ multiplications per position.
2. **Pointwise step:** A 1×1 convolution (pointwise convolution) then combines the

C_{in} intermediate maps into C_{out} output channels. Cost: $C_{\text{in}} \cdot C_{\text{out}}$ multiplications per position.

The total cost is $K^2 \cdot C_{\text{in}} + C_{\text{in}} \cdot C_{\text{out}}$ per position, compared to $K^2 \cdot C_{\text{in}} \cdot C_{\text{out}}$ for standard convolution. The reduction factor is approximately K^2, making depthwise convolution particularly effective for larger kernel sizes. This efficiency makes depthwise separable convolutions a key building block in lightweight architectures, as demonstrated by MobileNet [24] and adopted in subsequent YOLO variants for efficient feature extraction.

2.2.5 Optimizers

The choice of optimizer plays a critical role in training deep neural networks, affecting both convergence speed and final model quality. This section reviews two optimizers relevant to the proposed RayCastED model.

2.2.5.1 Stochastic Gradient Descent with Momentum

Stochastic Gradient Descent (SGD) remains one of the most widely used optimization algorithms in deep learning. Given a loss function $\mathcal{L}(\theta)$ parameterized by θ, SGD updates the parameters using the gradient of the loss estimated from a mini-batch of training samples:

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta \mathcal{L}(\theta_t) \tag{2-16}$$

where η is the learning rate. While simple, vanilla SGD can converge slowly in directions with low curvature and oscillate in directions with high curvature.

Momentum [25] addresses this by accumulating a velocity vector that smooths the gradient trajectory:

$$v_{t+1} = \mu v_t + \nabla_\theta \mathcal{L}(\theta_t), \quad \theta_{t+1} = \theta_t - \eta v_{t+1} \tag{2-17}$$

where $\mu \in [0, 1)$ is the momentum coefficient (typically $\mu = 0.9$ or $\mu = 0.99$). The velocity term accelerates convergence in consistent gradient directions and dampens oscillations, significantly improving training stability.

2.2.5.2 Muon Optimizer

The Muon optimizer [26] is a recent optimization algorithm designed for training neural networks with matrix-structured parameters. Muon applies Newton-Schulz orthogonalization to the gradient matrix at each step, implicitly constraining the spectral norm of

the weight matrices during training. This spectral normalization has been shown to improve training stability and generalization [26]. Muon has demonstrated strong empirical performance in training vision and language models, achieving faster convergence than AdamW while maintaining comparable or better final accuracy.

2.2.6 Discrete Wavelet Transform

2.2.6.1 Wavelet Decomposition Fundamentals

The wavelet transform provides a time-frequency (or space-frequency) analysis of a signal by decomposing it into components localized in both the spatial and frequency domains. Unlike the Fourier transform, which projects a signal onto infinite sinusoids and loses all spatial localization, the wavelet transform uses basis functions of finite support that capture both *where* and *at what scale* features occur.

The **Continuous Wavelet Transform (CWT)** of a signal $f(t) \in L^2(\mathbb{R})$ is defined with respect to a *mother wavelet* $\psi(t)$ [27]:

$$W_f(a,b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} f(t)\, \psi^*\left(\frac{t-b}{a}\right) dt \tag{2-18}$$

where $a \in \mathbb{R} \setminus \{0\}$ is the scale (dilation) parameter, $b \in \mathbb{R}$ is the translation (shift) parameter, and ψ^* denotes the complex conjugate. The factor $|a|^{-1/2}$ ensures energy normalization across scales. Small values of a produce narrow, high-frequency wavelets that capture fine details, while large values of a produce wide, low-frequency wavelets that capture coarse structures. By varying both a and b, the CWT produces a two-dimensional representation of the signal in the scale-translation plane, revealing how frequency content evolves across spatial position.

The CWT suffers from redundancy: the transform of a one-dimensional signal yields a two-dimensional coefficient map, and neighboring scales and translations are highly correlated. For computational efficiency and practical signal processing, the wavelet transform is commonly discretized.

2.2.6.2 Discrete Wavelet Transform

The **Discrete Wavelet Transform (DWT)** replaces the continuous scale-translation parameters with discrete samples, most commonly on a dyadic grid where $a = 2^{-j}$ and $b = k \cdot 2^{-j}$ for integers $j, k \in \mathbb{Z}$ [28]. Under a multiresolution analysis framework [28], the DWT can be efficiently implemented as a cascade of filter banks. At each level, the signal is convolved with a pair of quadrature mirror filters: a low-pass filter $h[n]$ associated with the scaling function $\phi(t)$, and a high-pass filter $g[n]$ associated with the wavelet function $\psi(t)$, followed by downsampling by a factor of 2:

$$c_{j+1}[k] = \sum_n h[n-2k]\, c_j[n], \qquad d_{j+1}[k] = \sum_n g[n-2k]\, c_j[n] \tag{2-19}$$

where c_j are the approximation coefficients at level j and d_j are the detail coefficients. The approximation c_{j+1} captures the low-frequency content at coarser resolution, while the detail d_{j+1} captures the high-frequency information lost during downsampling.

For a 2D signal such as an image, the DWT is applied separably: first filtering rows, then filtering columns of the intermediate result. This produces four subbands per decomposition level:

- **LL** (Low-Low): Approximation coefficients, obtained by applying the low-pass filter h to both rows and columns. This subband is a downsampled version of the input that preserves the overall spatial layout.
- **LH** (Low-High): Horizontal detail coefficients, obtained by applying h to rows and g to columns, capturing vertical edges.
- **HL** (High-Low): Vertical detail coefficients, obtained by applying g to rows and h to columns, capturing horizontal edges.
- **HH** (High-High): Diagonal detail coefficients, obtained by applying g to both rows and columns, capturing diagonal edges and fine textures.

Each subband has half the spatial resolution of the input in both dimensions. The decomposition can be applied recursively to the LL subband to obtain multi-level wavelet representations, where each level captures features at a different spatial scale.

The key advantage of wavelet decomposition over standard downsampling (e.g., strided convolution or max pooling) is that it provides a principled frequency separation. Standard downsampling discards high-frequency information entirely, which can eliminate fine textures and edges critical for accurate boundary delineation. Wavelet decomposition preserves high-frequency details in the LH, HL, and HH subbands, making these details available for subsequent processing.

2.2.6.3 Wavelet-Based Downsampling in CNNs

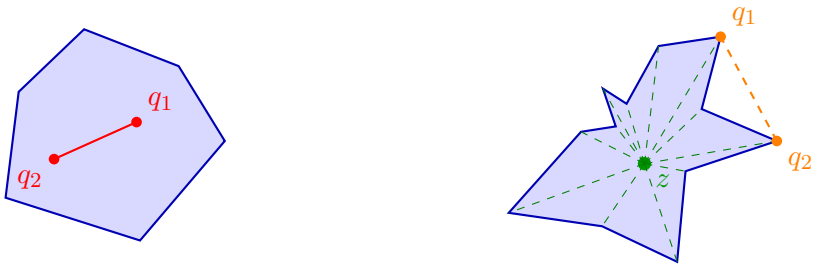
Integrating DWT into convolutional neural networks replaces standard downsampling operations (strided convolution or max pooling) with wavelet-based downsampling. In this approach, a feature map at one resolution is decomposed into its four wavelet subbands using DWT. The LL subband serves as the downsampled feature map for the next network stage, while the detail subbands (LH, HL, HH) can be concatenated with or injected into intermediate layers to preserve high-frequency information.

The wavelet-based downsampling operation can be expressed as follows. Given a feature map $\mathbf{F} \in \mathbb{R}^{H \times W \times C}$, the DWT produces four subbands each of size $\frac{H}{2} \times \frac{W}{2} \times C$. The LL subband $\mathbf{F}_{\text{LL}}$ is passed to the next convolutional block as the primary feature representation. The detail subbands can be processed by lightweight convolutional layers and merged back into the feature pipeline through skip connections or attention mechanisms.

This design is particularly relevant for nuclei detection, where cell boundaries are defined by fine high-frequency edges in the H&E image. By preserving these details through the downsampling hierarchy, wavelet-based networks can maintain boundary-aware features at deeper layers where the detection head operates. WaveConvX [29] demonstrates this principle in the context of histopathology image classification, showing that wavelet integration improves feature quality for FCN-based architectures.

2.2.7 Star-Convex Object Representation

2.2.7.1 Geometric Definitions



(a) Convex polygon

(b) Star-shaped polygon (non-convex)

Figure 2.1. Illustration of convex and star-shaped polygons. (a) A convex polygon: the segment $\overline{q_1 q_2}$ connecting any two interior points lies entirely within the polygon. (b) A star-shaped polygon that is not convex: the kernel point z (near the center) sees all boundary vertices (dashed green), yet the segment $\overline{q_1 q_2}$ connecting two interior points exits the polygon, demonstrating that the interior is not a convex set.

A **convex polygon** is a polygon whose interior forms a convex set. A polygon P is convex if its interior region is a convex set [30]. A region R is a convex set if, for any two points q_1 and q_2 in R, the segment $\overline{q_1 q_2}$ is entirely contained in R [30]. Intuitively, for a convex polygon, the line segment connecting any two points inside the polygon lies entirely within the polygon, and no interior angle exceeds $180°$.

A **star-shaped polygon** generalizes the notion of convexity. A polygon P is star-shaped if there exists a point z not external to P such that for all points p of P, the line segment $\overline{zp}$ lies entirely within P [30]. The set of all such points z is called the *kernel* of the polygon. A star-shaped polygon can be viewed as one where there is at least one interior point visible to every other point in the polygon. Every convex polygon is star-

shaped (the kernel is the entire interior), but a star-shaped polygon may have indentations that make it non-convex.

A **star-convex polygon** is a polygon P that is simultaneously star-shaped and convex. This dual property makes star-convex polygons particularly useful for representing cell nuclei: because they are convex, every boundary point can be uniquely described by a single ray emanating from the centroid, and because they are star-shaped, the centroid can be placed at a kernel point where all such rays intersect the boundary exactly once. In polar coordinates, a star-convex polygon centered at its centroid $\mathbf{c}$ can therefore be completely represented by a set of radial distances $\{d_0, d_1, \ldots, d_{n-1}\}$ measured at n fixed angular intervals from the centroid to the boundary.

2.2.7.2 Suitability for Nuclei Representation

The star-convex polygon representation is well-suited for representing cell nuclei because of their characteristic morphology. The majority of cell nuclei encountered in H&E-stained tissue sections are approximately convex or roundish in shape [9]. Healthy epithelial nuclei are round or elliptical, fibroblasts and endothelial cells are elongated but convex, and even malignant nuclei with pleomorphic variations largely retain a convex silhouette in their two-dimensional projection. This near-ubiquitous convexity means that a star-convex polygon can capture the full boundary geometry of a nucleus with a compact set of radial distances, avoiding the information loss of axis-aligned bounding boxes while remaining far more parameter-efficient than dense per-pixel segmentation masks [8].

2.2.8 End-to-End Detection with Bipartite Matching

2.2.8.1 Bipartite Matching: Hungarian Algorithm

Bipartite matching establishes a one-to-one correspondence between two sets of elements by finding an optimal assignment that minimizes a total cost. In the context of object detection, bipartite matching assigns each ground-truth object to exactly one prediction and each prediction to at most one ground-truth object, eliminating the ambiguity inherent in IoU-based assignment where multiple predictions can match the same ground-truth object.

The assignment is formulated as a minimum-cost bipartite matching problem. Given a set of N ground-truth objects and M predictions, a cost matrix $\mathbf{C} \in \mathbb{R}^{N \times M}$ is constructed where $C_{i,j}$ measures the dissimilarity between ground-truth i and prediction j. A common choice is the negative IoU: $C_{i,j} = 1 - \text{IoU}(g_i, p_j)$. The optimal assignment is found by solving:

$$
\begin{aligned}
\min_{x_{i,j}} \quad & \sum_{i=1}^{N} \sum_{j=1}^{M} C_{i,j}\, x_{i,j} \\
\text{s.t.} \quad & \sum_{j=1}^{M} x_{i,j} = 1, \quad \forall i \in \{1, \ldots, N\} \\
& \sum_{i=1}^{N} x_{i,j} \leq 1, \quad \forall j \in \{1, \ldots, M\} \\
& x_{i,j} \in \{0, 1\}
\end{aligned}
\tag{2-20}
$$

where $x_{i,j} = 1$ if ground-truth i is assigned to prediction j, and 0 otherwise. The first constraint ensures every ground-truth object receives exactly one match. The second constraint ensures no prediction is matched to more than one ground-truth object. When $M > N$, predictions not assigned to any ground-truth are treated as background.

The **Hungarian algorithm** (Kuhn–Munkres) solves this problem in $O(\max(N, M)^3)$ time through the following steps on a square cost matrix (padded with large values if $N \neq M$): (1) subtract the row minimum from each row, (2) subtract the column minimum from each column, (3) cover all zero entries with the minimum number of horizontal and vertical lines, (4) if the number of lines equals N, an optimal assignment is found by selecting independent zeros; otherwise, subtract the smallest uncovered value from all uncovered entries, add it to all doubly covered entries, and repeat from step 3.

This bipartite matching formulation is the foundation of end-to-end detection frameworks such as DETR. By guaranteeing a unique assignment, it eliminates the need for NMS as a post-processing step, since each ground-truth object is matched to exactly one prediction and there are no duplicate assignments to suppress.

2.2.8.2 Task-Aligned Matching

Task-aligned matching is an assignment strategy used in YOLO-based detection frameworks that jointly considers classification and localization quality when matching predictions to ground-truth objects. Rather than using IoU alone, task-aligned matching defines a combined alignment metric:

$$
t = s^{\alpha} \cdot u^{\beta} \tag{2-21}
$$

where s is the classification confidence score, u is the predicted IoU between the prediction and the ground-truth object, and α, β are weighting hyperparameters that balance the contribution of classification and localization. A prediction is assigned to a ground-truth object only if its alignment metric t exceeds a threshold, and each ground-truth object is assigned to the prediction with the highest t value. This ensures that

predictions assigned to an object are both well-localized (high IoU) and confidently classified, improving training stability.

2.2.8.3 Dual-Label Assignment

Dual-label assignment, introduced in YOLO26, is a training strategy that combines one-to-many and one-to-one matching within a single architecture to achieve the benefits of both paradigms. The model employs two parallel detection heads: a **one-to-many head** that assigns multiple predictions to each ground-truth object (as in conventional YOLO training), and a **one-to-one head** that enforces a unique bipartite matching between predictions and ground-truth objects (as in DETR-style training).

The two heads are balanced through an **annealing weight schedule** that controls their relative contribution to the total loss. During early training epochs, the one-to-many head dominates, providing dense gradient signals that stabilize training and accelerate feature learning. As training progresses, the weight shifts gradually toward the one-to-one head, which refines the predictions to produce unambiguous, duplicate-free detections. This transition allows the model to benefit from the richer supervision of one-to-many matching early on while converging to the cleaner one-to-one assignment necessary for NMS-free inference. At inference time, only the one-to-one head is used, eliminating the need for non-maximum suppression.

2.2.9 Evaluation Metrics

2.2.9.1 Precision, Recall, and F1 Score

Object detection and segmentation are evaluated by comparing predicted instances against ground-truth annotations. Each prediction is classified as a **true positive (TP)** if it correctly matches a ground-truth object, a **false positive (FP)** if it does not correspond to any ground-truth object, or a **false negative (FN)** if a ground-truth object is missed. The definition of a correct match depends on the task: for bounding-box or polygon-based detection, a prediction is typically considered a TP if its intersection over union (IoU) with the ground truth exceeds a threshold τ (commonly $\tau = 0.5$); for centroid-based detection, a prediction may be considered a TP if the distance between the predicted and ground-truth centroids falls below a distance threshold.

Given the counts of TP, FP, and FN, precision and recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN} \tag{2-22}$$

Precision measures the fraction of predictions that are correct (fewer FPs yield higher precision), while recall measures the fraction of ground-truth objects that are found (fewer FNs yield higher recall). The two are often in tension: a model that

predicts conservatively achieves high precision but low recall, while a model that predicts aggressively achieves high recall but low precision.

The **F1 score** is the harmonic mean of precision and recall, providing a single balanced metric:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2-23}$$

2.2.9.2 Mean Average Precision (mAP)

Mean Average Precision (mAP) is the standard evaluation metric for object detection. It is computed by first calculating precision–recall curves for each object class and then averaging the area under these curves across all classes.

Given a set of detections ranked by confidence score and a chosen match criterion (typically IoU > 0.5), precision and recall are computed at each confidence threshold to produce a precision–recall curve. The Average Precision (AP) for a single class is then the area under the interpolated precision–recall curve:

$$\text{AP} = \int_0^1 P_{\text{interp}}(R)\, dR \tag{2-24}$$

where $P_{\text{interp}}(R) = \max_{\tilde{R} \geq R} P(\tilde{R})$ is the interpolated precision at recall level R. In practice, this is approximated by evaluating precision at a discrete set of recall points.

The mean Average Precision is then the mean of per-class AP values:

$$\text{mAP} = \frac{1}{C} \sum_{c=1}^{C} \text{AP}_c \tag{2-25}$$

where C is the number of object classes. When reported at a specific IoU threshold (e.g., mAP@0.5), detections are matched using that threshold. The COCO evaluation protocol averages mAP over multiple IoU thresholds:

$$\text{mAP@[.50:.05:.95]} = \frac{1}{K} \sum_{k=1}^{K} \text{mAP}(\theta_k) \tag{2-26}$$

where $\theta_k \in \{0.50, 0.55, \ldots, 0.95\}$ and $K = 10$. This provides a more comprehensive assessment of localization accuracy than evaluation at a single threshold. The notation mAP@0.50 (or AP_{50}) refers to evaluation at a single IoU threshold of 0.50.

2.2.9.3 Panoptic Quality (PQ)

Panoptic Quality (PQ) was introduced by Kirillov et al. [31] as a unified metric for panoptic segmentation, combining both recognition quality (detection accuracy) and segmentation quality (boundary accuracy). PQ has become the standard evaluation metric for nuclei instance segmentation benchmarks such as PanNuke [32] and CoNIC [33].

Given a ground truth segmentation G and a predicted segmentation P, each instance in G and P is uniquely labeled. A predicted segment $\hat{s}$ is matched to a ground truth segment s if and only if their intersection over union (IoU) exceeds a threshold, typically $\text{IoU} > 0.5$:

$$\text{IoU}(s, \hat{s}) = \frac{|s \cap \hat{s}|}{|s \cup \hat{s}|} \tag{2-27}$$

After matching, PQ is computed as:

$$\text{PQ} = \underbrace{\frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}}_{\text{Recognition Quality (RQ)}} \times \underbrace{\frac{\sum_{(s,\hat{s}) \in TP} \text{IoU}(s, \hat{s})}{|TP|}}_{\text{Segmentation Quality (SQ)}} \tag{2-28}$$

where TP is the set of true positive pairs (matched ground truth–prediction pairs), FP is the set of false positive predictions (unmatched predictions), and FN is the set of false negative ground truth segments (unmatched ground truth). PQ thus decomposes into two interpretable components:

- **Recognition Quality (RQ):** measures detection accuracy and is equivalent to the F1 score computed over matched and unmatched segments.
- **Segmentation Quality (SQ):** measures the average IoU of correctly matched pairs, quantifying how precisely the predicted boundaries align with the ground truth.

Several PQ variants are commonly reported in nuclei segmentation literature:

- **Instance PQ (iPQ):** evaluates PQ on a per-image basis and then averages across all images in the dataset. Each image contributes equally regardless of the number of nuclei it contains.
- **Multi-class PQ (mPQ):** extends PQ to multi-class settings by computing PQ independently for each nuclei type and then averaging across all classes:

$$\text{mPQ} = \frac{1}{C} \sum_{c=1}^{C} \text{PQ}_c \tag{2-29}$$

where C is the number of classes and PQ_c is the PQ computed for class c only.

- **Binary PQ (bPQ):** computes PQ in a binary setting where all nuclei types are treated as a single class, ignoring type-specific distinctions. This is useful for evaluating pure instance segmentation performance independent of classification accuracy.
- **PanNuke PQ (nuPQ):** the specific PQ variant used in the PanNuke benchmark, which computes mPQ across the 19 nuclei type categories defined in the dataset.

2.2.9.4 Aggregated Jaccard Index (AJI)

The Aggregated Jaccard Index (AJI) [34] is an instance-level segmentation metric designed to address the overcounting problem in the standard Jaccard Index when applied to multi-object segmentation. In nuclei segmentation, the standard Jaccard Index (Intersection over Union, IoU) computed per instance and then averaged can overestimate performance because each ground-truth nucleus is free to match with any predicted nucleus independently, and unmatched predictions do not penalize the score.

AJI resolves this by computing a single global intersection-over-union across all instances. Given a set of ground-truth nuclei $G = \{g_1, g_2, \ldots, g_N\}$ and predicted nuclei $P = \{p_1, p_2, \ldots, p_M\}$, AJI first finds the optimal one-to-one assignment that maximizes total pairwise IoU. Let $\mathcal{M} \subseteq G \times P$ denote the set of matched pairs, $G_{\text{matched}} \subseteq G$ the set of matched ground-truth nuclei, and $P_{\text{matched}} \subseteq P$ the set of matched predicted nuclei. The AJI is then defined as:

$$\text{AJI} = \frac{\sum_{(g_i, p_j) \in \mathcal{M}} |g_i \cap p_j|}{\sum_{(g_i, p_j) \in \mathcal{M}} |g_i \cup p_j| + \sum_{g_k \in G \setminus G_{\text{matched}}} |g_k| + \sum_{p_l \in P \setminus P_{\text{matched}}} |p_l|} \tag{2-30}$$

The numerator sums the intersection area over all matched pairs, while the denominator sums the union area over all matched pairs plus the areas of all unmatched ground-truth and predicted nuclei. This formulation penalizes both false negatives (unmatched ground-truth nuclei) and false positives (unmatched predictions), providing a more stringent evaluation than per-instance averaging.

CHAPTER III
RESEARCH METHODOLOGY

3.1 Datasets and Environmental Setup

This study used publicly available secondary datasets for training and evaluation. Table 3.1 summarized the datasets and their roles.

Table 3.1. Summary of datasets used in this study.

Dataset	Purpose	Tissue	Format
PanNuke	Primary benchmarking	Multi-organ	Pickled bytes
PUMA	TILs detection	Melanoma	TIFF, GeoJSON
panopTILs	TILs detection	Breast	PNG
MoNuSAC	TILs detection	Multi-organ	Pickled bytes

3.1.1 Datasets

3.1.1.1 PanNuke

PanNuke [32, 35] is a large-scale pan-cancer nucleus instance segmentation dataset containing approximately 189,744 labeled nuclei across 19 tissue types. Each nucleus is annotated with an instance mask and one of five nuclear categories. The dataset is provided as pickled bytes containing image patches and corresponding label masks. In this study, PanNuke served as the primary benchmarking dataset for evaluating nuclei detection and segmentation performance.

3.1.1.2 PUMA

PUMA (Melanoma Histopathology Dataset with Tissue and Nuclei Annotations) [36] provides annotated histopathology images for TILs detection in melanoma tissue. The dataset is distributed as TIFF images paired with GeoJSON annotation files containing polygon-level cell boundary annotations. This dataset was used to evaluate RayCastED's ability to detect and delineate TILs in a clinically relevant setting.

3.1.1.3 panopTILs

panopTILs [37] provides TILs annotations on breast cancer histopathology images. The dataset is distributed as PNG image patches with corresponding annotation files. This dataset supplemented the TILs evaluation alongside PUMA, providing breast tissue samples for assessing TILs detection performance across cancer types.

3.1.1.4 MoNuSAC

MoNuSAC [38] provides annotated nuclei across multiple organ types with four nuclear categories: epithelial, lymphocyte, neutrophil, and macrophage. The dataset is provided as pickled bytes. The lymphocyte annotations in MoNuSAC were particularly relevant for evaluating TILs detection, as lymphocytes are the primary cell type of interest in TILs scoring.

3.1.2 Environmental Setup

3.2 Methods

3.2.1 Data Preprocessing

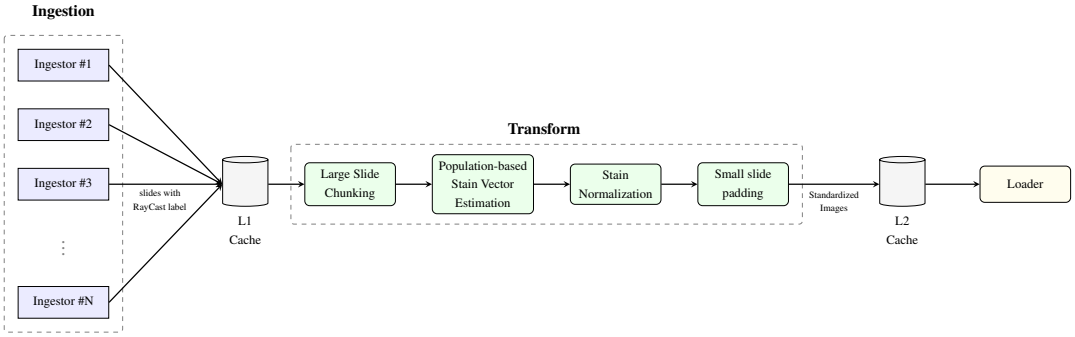


Figure 3.2. High-level Pipeline Design

The data preprocessing pipeline consists of three sequential stages: Ingestion, Transform, and Loader. The Ingestion stage reads data from various source formats and normalizes them into a consistent internal representation. The Transform stage applies standardization and normalization operations to produce uniformly preprocessed samples. The Loader stage applies online augmentations and feeds the prepared data to the training loop. Intermediate results are cached between stages to isolate failures and avoid redundant recomputation.

3.2.1.1 Ingestion Pipeline

3.2.1.1.1 RayCast Representation.

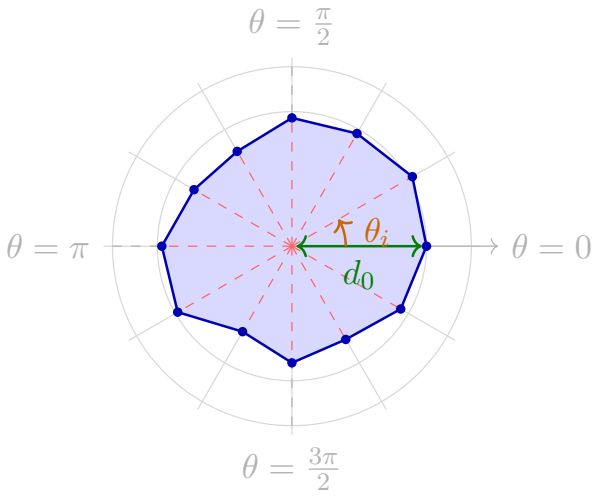


Figure 3.3. A star-convex polygon with centroid $\mathbf{c}$, polar grid, and 12 radial rays with angular spacing θ_i.

The RayCast representation models a nucleus as a star-convex polygon with a fixed angular resolution. A star-convex polygon is defined by a centroid $\mathbf{P}$ and the radial distances to its boundary along R uniformly spaced angular directions, as illustrated in Figure 3.3. Unlike standard polygon representations that store an arbitrary number of contour vertices, the RayCast representation normalizes every nucleus to a fixed-length vector:

$$\mathbf{a} = (c,\ c_x,\ c_y,\ d_0,\ d_1,\ \ldots,\ d_{R-1})^\top \in \mathbb{R}^{3+R} \tag{3-1}$$

where c is the class label, (c_x, c_y) is the centroid, R is the number of rays (set to 32 in this study and constrained to be a multiple of 4), and d_r is the Euclidean distance from the centroid to the polygon boundary along ray r at angle $\theta_r = r \cdot 2\pi/R$. The angular convention is $\theta_0 = 0$ (east, $+X$ axis), increasing counter-clockwise. This representation is fully reversible: given the centroid and radial distances, the polygon can be reconstructed by placing vertices at $\mathbf{V}_r = \mathbf{P} + d_r \cdot \mathbf{D}_r$ and connecting them cyclically, where $\mathbf{D}_r = (\cos\theta_r,\ \sin\theta_r)^\top$.

The fixed-length structure is essential for deep learning: it allows the network to predict polygon parameters as a regression task with a constant number of output channels, independent of the nuclear shape complexity. The star-convex constraint ensures that the resulting polygon is always simple and topologically valid, unlike unconstrained point sequences that may produce self-intersecting or degenerate boundaries.

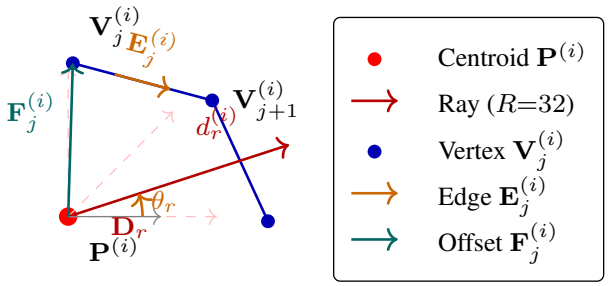


Figure 3.4. RayCast representation of a cell polygon. From the area-weighted centroid $\mathbf{P}$, $R = 32$ rays are cast at uniformly spaced angles. Each distance d_r records the nearest boundary intersection along ray r.

The generation pipeline assumes it receives a standard polygon $\mathcal{P}$ defined by an ordered set of K vertices $\{\mathbf{V}_j\}_{j=0}^{K-1}$ in $\mathbb{R}^2$, where the specific source format (binary masks, GeoJSON coordinates, or integer label maps) has been resolved by the dataset-specific ingestor described in Section 3.2.1.1.2. The generation proceeds through several steps.

Raster Mask to Contour Extraction.

Datasets such as PanNuke and MoNuSAC provide per-cell binary raster masks rather than polygon coordinates. For each cell, the binary mask is decoded and contour vertices are extracted via Moore-neighbor tracing (OpenCV's `findContours` with `CHAIN_APPROX_NONE`) to preserve full boundary fidelity. When a mask produces multiple contours, only the contour with the largest enclosed area is retained. Contours with fewer than 3 vertices are discarded.

Centroid Computation.

The area-weighted centroid serves as the origin from which all rays emanate. Given the polygon vertices with cyclic indexing $\mathbf{V}_K \equiv \mathbf{V}_0$, the signed area is computed via the shoelace formula:

$$A = \frac{1}{2} \sum_{j=0}^{K-1} (x_j \, y_{j+1} - x_{j+1} \, y_j) \tag{3-2}$$

Polygons with $|A| < \varepsilon$ (where $\varepsilon = 10^{-12}$) are near-zero area; in this case the centroid is set to the first vertex $\mathbf{V}_0$. The area-weighted centroid coordinates are:

$$c_x = \frac{1}{6A} \sum_{j=0}^{K-1} (x_j + x_{j+1}) \, (x_j \, y_{j+1} - x_{j+1} \, y_j) \tag{3-3}$$

$$c_y = \frac{1}{6A} \sum_{j=0}^{K-1} (y_j + y_{j+1}) \, (x_j \, y_{j+1} - x_{j+1} \, y_j) \tag{3-4}$$

Ray–Edge Intersection.

The core computation solves for the intersection of each ray with every polygon edge. The ray from centroid $\mathbf{P}$ in unit direction $\mathbf{D}_r = (\cos\theta_r, \, \sin\theta_r)^\top$ is parameterized as:

$$\mathbf{L}_r(t) = \mathbf{P} + t \, \mathbf{D}_r, \quad t \geq 0 \tag{3-5}$$

An edge segment from $\mathbf{V}_j$ to $\mathbf{V}_{j+1}$ with edge vector $\mathbf{E}_j = \mathbf{V}_{j+1} - \mathbf{V}_j$ is:

$$\mathbf{Q}_j(s) = \mathbf{V}_j + s \, \mathbf{E}_j, \quad s \in [0, 1] \tag{3-6}$$

Both equations describe the same geometric primitive – a line – differing only in their parameter bounds. The ray is unbounded on one side ($t \geq 0$) and the edge segment is bounded on both sides ($s \in [0, 1]$). Their intersection reduces to solving two linear equations.

Setting $\mathbf{L}_r(t) = \mathbf{Q}_j(s)$ and defining the offset $\mathbf{F}_j = \mathbf{V}_j - \mathbf{P}$ yields a 2×2 linear system:

$$\underbrace{\begin{bmatrix} \cos\theta_r & -E_{j,x} \\ \sin\theta_r & -E_{j,y} \end{bmatrix}}_{\mathbf{M}_{rj}} \begin{bmatrix} t \\ s \end{bmatrix} = \underbrace{\begin{bmatrix} F_{j,x} \\ F_{j,y} \end{bmatrix}}_{\mathbf{b}_j} \tag{3-7}$$

The system is solved via Cramer's rule using the 2D cross product $\mathbf{a} \times \mathbf{b} = a_x b_y - a_y b_x$:

$$\det_{rj} = \det(\mathbf{M}_{rj}) = -\cos\theta_r \cdot E_{j,y} + \sin\theta_r \cdot E_{j,x} \tag{3-8}$$

$$t_{rj} = \frac{-\mathbf{F}_j \times \mathbf{E}_j}{\det_{rj}}, \qquad s_{rj} = \frac{\mathbf{D}_r \times \mathbf{F}_j}{\det_{rj}} \tag{3-9}$$

The right-hand side $\mathbf{b}_j$ is independent of the ray direction; only the first column of $\mathbf{M}_{rj}$ varies with r.

Validity Filtering and Nearest-Hit Selection.

An intersection is valid if and only if: (1) $|\text{det}_{rj}| > \epsilon_d$ (ray is not parallel to the edge, $\epsilon_d = 10^{-10}$), (2) $t_{rj} > 0$ (intersection lies in front of the centroid; the strict inequality prevents self-intersection artifacts when the centroid lies exactly on the boundary), (3) $-\epsilon \leq s_{rj} \leq 1 + \epsilon$ (intersection falls on the edge segment, $\epsilon = 10^{-6}$), and (4) the edge is a real polygon edge. Degenerate cases are handled as follows:

- **Ray through shared vertex:** both adjacent edges produce valid hits with the same t; the minimum selection naturally resolves this.
- **Ray tangent to edge:** filtered by the determinant threshold $|\text{det}_{rj}| < \epsilon_d$.
- **Ray parallel but offset:** produces $t \to \pm\infty$, filtered by the $t > 0$ condition.
- **Ray collinear with edge:** the edge lies along the ray direction; d_r is conservatively set to $d_r = 0$.

For each ray r, the boundary distance is the minimum positive t among valid intersections:

$$d_r = \min_{j \in \mathcal{V}_r} t_{rj}, \qquad \mathcal{V}_r = \{j : \text{valid}_{rj} = \text{True}\} \tag{3-10}$$

Since $\mathbf{D}_r$ is a unit vector, $d_r = t_{rj^*}$ is directly the Euclidean distance from the centroid to the boundary intersection. Rays with no valid intersection ($\mathcal{V}_r = \emptyset$) are assigned $d_r = 0$, following the convention that zero distance indicates a missed ray.

Decoding.

The raycast representation decodes back to a polygon by reconstructing vertex positions from the centroid and radial distances:

$$\mathbf{V}_r = \mathbf{P} + d_r \cdot \mathbf{D}_r, \qquad r = 0, \ldots, R-1 \tag{3-11}$$

connected cyclically. The angular direction vectors $\mathbf{D}_r$ are precomputed once and shared across ingestion, model inference, loss computation, and evaluation.

3.2.1.1.2 Data Ingestion Design.

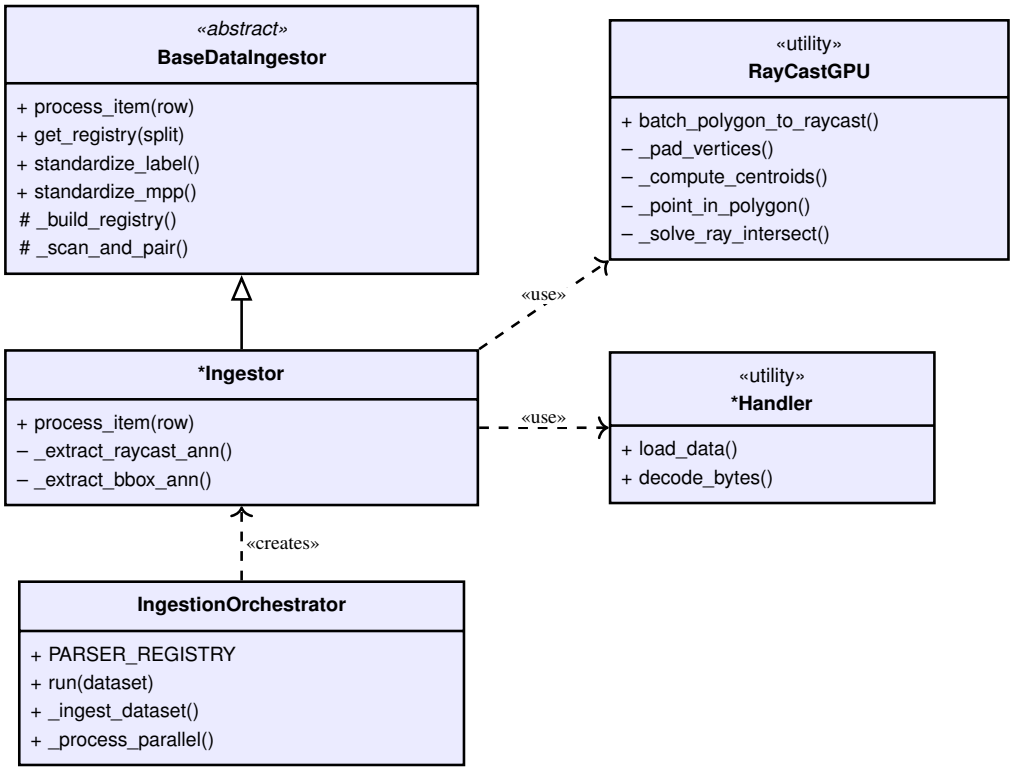


Figure 3.5. UML class diagram of the modular ingestion pipeline. `BaseDataIngestor` defines the shared infrastructure; concrete ingestors implement format-specific `process_item` logic; all converge on the unified raycast annotation format via `RayCastGPU`.

A practical computational pathology system must ingest annotations from heterogeneous source formats: binary instance masks, polygon coordinates, comma-separated coordinate strings, and integer label maps. Rather than implementing a monolithic converter per dataset, the ingestion pipeline follows a modular architecture with three design principles.

First, **format-agnostic core logic** (centroid computation, ray–edge intersection solving, MPP harmonization, and label translation) is shared across all ingestors. Second, each concrete ingestor implements only the **format-specific extraction** logic that parses its source format into the common intermediate representation: polygon vertex arrays with class labels. Third, **configuration-driven routing** specifies the choice of ingestor, split strategy, modality pairing, and label mapping declaratively, without code changes.

The architecture is built around an abstract base class that defines the contract every ingestor must fulfill: for each region of interest, produce a unified output $(id_{\text{roi}}, \mathbf{I}, \mathbf{A}, t_{\text{tissue}})$ consisting of an identifier, an RGB image, a raycast annotation matrix $\mathbf{A} \in \mathbb{R}^{N\times(3+R)}$, and a tissue provenance label. The sole abstract method is `process_item`, which each concrete ingestor implements for its specific format.

The base class provides a template method for **registry construction** in three phases: (1) file discovery and pairing, where image files are matched with annotation files using either bundled-archive (same file) or physical-parallel (paired directories) strategies; (2) split assignment, where each pair is assigned to train/val/test using physical subdirectory names, filename regex patterns, or stratified random sampling by tissue type; and (3) registry assembly into a work queue. A **two-step label translation** decouples source vocabularies from the global integer encoding: a namespace map $\mathcal{N}$ translates raw labels to standardized strings shared across all datasets, and a global cell map $\mathcal{G}$ maps standardized strings to integer class IDs. Adding a new dataset requires only a new namespace map. Similarly, all annotations are rescaled to a common target microns-per-pixel via **MPP harmonization**, scaling image dimensions and annotation coordinates uniformly by the ratio $\rho = \sigma_{\text{native}}/\sigma_{\text{target}}$.

Concrete ingestors differ only in how they extract polygon vertices: mask-based sources decode compressed byte strings via Moore-neighbor contour tracing, while vector-based sources read polygon coordinates directly from geometry arrays or comma-separated coordinate strings requiring no rasterization. Mask-based ingestors process ROIs in parallel using a process pool, deferring GPU ray-cast conversion to the main process to avoid CUDA cross-process deadlock. All ingestors converge through a single conversion entry point, `RayCastGPU.batch_polygon_to_raycast`, which receives the format-independent intermediate representation (variable-length vertex arrays and class labels) and produces the unified raycast annotation $\mathbf{A}$.

The `IngestionOrchestrator` governs the end-to-end process: a string-keyed registry maps configuration keys to concrete ingestor classes, validated at both config-load time and runtime. A two-layer YAML hierarchy (global defaults overridden per dataset) enables declarative specification of all parameters. Row-level errors are caught and logged without terminating the remaining dataset, so a single corrupt ROI does not halt ingestion. This separation of concerns means that adding a new dataset typically requires only a YAML configuration entry with a namespace map and format key, while the computational core remains invariant across all sources.

3.2.1.2 Transform Pipeline

3.2.1.2.1 Data Profile Registration.

Before any per-tile transformation is applied, the ingestion pipeline registers a data profile for each image in the dataset. This profile captures three categories of metadata that downstream transforms consume: (1) **tissue type and dataset source**, recorded as a string identifier and integer encoding, enabling stratified sampling across datasets and tissue categories during population-level stain estimation; (2) **native image dimensions** (W, H) and **microns-per-pixel** (MPP), required by the size standardization

step to compute chopping and padding parameters; and (3) **tile-level stain statistics**, including a boolean flag indicating whether the tile contains sufficient tissue content (at least 100 non-white pixels) to be eligible for stain vector estimation. Profiles are assembled during the ingestion registry phase and serialized alongside the raycast annotations, so that the transform pipeline can load them without re-reading the source images. This separation ensures that the transform pipeline operates purely on pre-computed metadata rather than performing redundant I/O.

3.2.1.2.2 Stain Vector Estimation and Normalization.

Histopathology images exhibit substantial color variation across scanners, staining batches, and laboratories. A model trained on darkly-stained slides may perform poorly on lightly-stained ones. The transform pipeline addresses this through a three-phase process. First, images are chopped into $S \times S$ tiles. Second, a subset of tiles is Pareto-sampled, the Macenko method [19] estimates per-tile stain vectors, and the results are aggregated into a population-level profile. Third, every tile is normalized against this profile and padded to a uniform size.

Population-Level Stain Vector Estimation.

Rather than processing every tile – which wastes computation and allows large datasets to dominate – stratified Pareto sampling is applied following Agraz et al. [39], who showed that 20% of a cohort suffices for a representative profile. The chunked registry is partitioned into strata by dataset and tissue type, ensuring each dataset contributes proportionally and rare tissue types are represented. Within each stratum containing n_j tiles, the number sampled is:

$$k_j = \max(1,\ \lceil 0.20 \cdot n_j \rceil) \tag{3-12}$$

Tiles are drawn uniformly at random with a fixed seed. The estimation set $\mathcal{S}$ is the union across all J strata.

For each sampled tile $k \in \mathcal{S}$, the Macenko method estimates a per-tile stain matrix $\mathbf{M}_{\text{stain}}^{(k)} \in \mathbb{R}^{2\times 3}$ and maximum concentrations $\mathbf{c}_{\max}^{(k)} \in \mathbb{R}^2$. The population profile is aggregated across the K tiles in $\mathcal{S}$ that yield valid estimates. The stain matrix is averaged since stain directions vary smoothly, while maximum concentrations use the 99th percentile to establish a robust upper bound:

$$\overline{\mathbf{M}} = \frac{1}{K}\sum_{k\in\mathcal{S}} \mathbf{M}_{\text{stain}}^{(k)}, \qquad \mathbf{c}_{\max}^{*} = P_{99}\big(\{\mathbf{c}_{\max}^{(k)}\}_{k\in\mathcal{S}}\big) \tag{3-13}$$

Tiles that fail estimation (all-white or fewer than 100 tissue pixels) are silently excluded from K. The profile is serialized as JSON containing $\overline{\mathrm{M}}$, $\mathbf{c}^*_{\max}$, the estimation method, and the tile count K for reuse during normalization and at inference time.

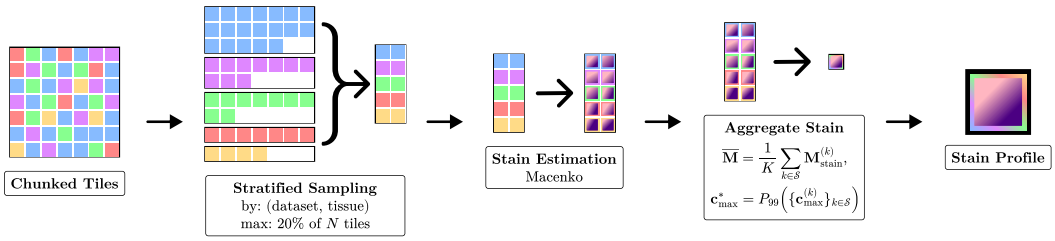


Figure 3.6. Population-level stain vector estimation. A stratified 20% subset of tiles is sampled per dataset and tissue type; the Macenko method estimates per-tile stain matrices and maximum concentrations, which are aggregated into the population profile $(\overline{\mathrm{M}},\ \mathbf{c}^*_{\max})$.

Per-Tile Stain Normalization.

Given the population target $(\overline{\mathrm{M}},\ \mathbf{c}^*_{\max})$ and a per-tile source estimate $(\mathbf{M}_{\text{source}},\ \mathbf{c}_{\text{source}})$, normalization consists of five steps.

The tile is first converted from RGB to optical density space via the Beer–Lambert law:

$$\mathrm{OD} = -\log_{10}\left(\frac{I_o + 1}{I_o}\right) \tag{3-14}$$

where the $+1$ avoids $\log(0)$ at black pixels. The OD image is then decomposed into Hematoxylin and Eosin stain concentration maps using the pseudoinverse of the source stain matrix:

$$\mathbf{C}_{\text{source}} = \mathrm{OD} \cdot \mathbf{M}^{+}_{\text{source}} \tag{3-15}$$

Each channel's concentration is rescaled to match the population target range by the per-channel percentile ratio:

$$\mathbf{C}_{\text{norm}} = \mathbf{C}_{\text{source}} \odot \frac{\mathbf{c}^*_{\max}}{\mathbf{c}_{\text{source}}} \tag{3-16}$$

A small epsilon (10^{-6}) prevents division by zero when a concentration channel is empty. The normalized OD is reconstructed using the population mean stain directions, replacing the source-specific color signature with the standardized one:

$$\text{OD}_{\text{norm}} = \mathbf{C}_{\text{norm}} \cdot \overline{\mathbf{M}} \tag{3-17}$$

Finally, the normalized OD is converted back to RGB and clipped to $[0, 255]$:

$$I_{\text{norm}} = I_o \cdot 10^{-\text{OD}_{\text{norm}}} - 1 \tag{3-18}$$

The full transform can be expressed compactly as:

$$I_{\text{norm}} = I_o \cdot 10^{-\left(\text{OD}\cdot\mathbf{M}^{+}_{\text{source}} \odot \frac{\mathbf{c}^{*}_{\max}}{\mathbf{c}_{\text{source}}}\right)\cdot\overline{\mathbf{M}}} - 1 \tag{3-19}$$

3.2.1.2.3 Image Size Standardization.

Every ingested image must be reduced to a fixed $S \times S$ canvas (defaulting to 256×256 pixels) suitable for batched GPU training. Two operations handle the two possible regimes: images larger than S in either dimension are **chopped** into overlapping tiles; images smaller than or equal to S are **padded** with white pixels at the bottom and right edges.

Chopping: Dynamic Overlap Distribution.

For an image dimension $L > S$ with minimum overlap fraction α_{overlap}, the number of tiles n is computed from the effective stride $\sigma = S - \lfloor S \cdot \alpha_{\text{overlap}} \rfloor$. Tile starting positions are distributed uniformly via linspace$(0, L - S, n)$, which distributes any excess overlap evenly across all seams rather than concentrating it at the boundary.

Annotation-Aware Raycast Splitting.

Each cell is assigned to exactly one tile via a centroid-owns-cell policy: a cell belongs to the tile containing its centroid (c_x, c_y). The splitting procedure has four steps. First, centroids are filtered to retain only cells within the tile bounds. Second, centroid coordinates are shifted to crop-relative space: $c'_x = c_x - x_0$, $c'_y = c_y - y_0$. Third, for each of the R rays, the distance to each of the four tile boundaries is computed. For a ray in direction $\mathbf{D}_r = (\cos\theta_r, \sin\theta_r)$, the boundary distances are:

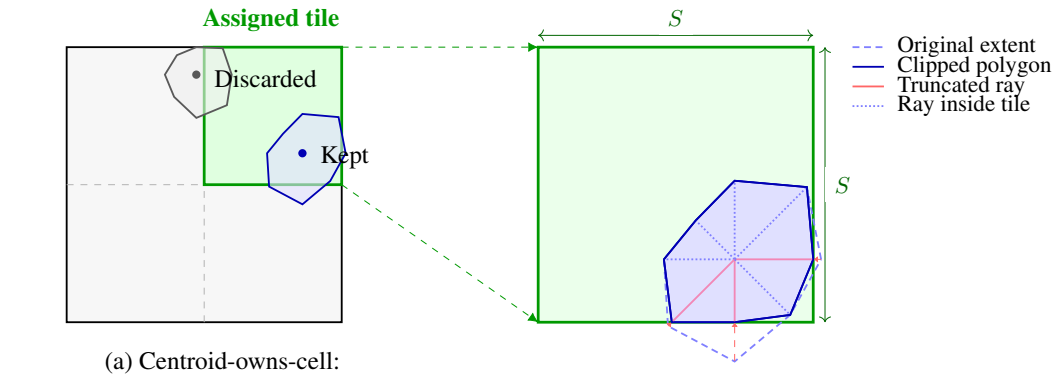
$$d_{\text{right}} = \frac{w - c'_x}{\cos\theta_r}, \quad d_{\text{left}} = \frac{-c'_x}{\cos\theta_r}, \quad d_{\text{bottom}} = \frac{h - c'_y}{\sin\theta_r}, \quad d_{\text{top}} = \frac{-c'_y}{\sin\theta_r} \tag{3-20}$$

where terms with $|\cos\theta_r| \leq \varepsilon$ or $|\sin\theta_r| \leq \varepsilon$ are set to $+\infty$ (the ray is parallel to that boundary). The maximum permissible distance is $d_{\max} = \min(d_{\text{right}}, d_{\text{left}}, d_{\text{bottom}}, d_{\text{top}})$,

and each ray is clipped: $d'_r = \min(d_r, d_{\max})$. Fourth, cells that lose too many rays to boundary clipping are discarded via a survival rate filter:

$$\frac{|\{r : d'_r > 0\}|}{R} \geq \tau_{\text{survival}} \tag{3-21}$$

where $\tau_{\text{survival}} = 0.5$. This prevents heavily truncated cells with unreliable shape representations from entering the training data. Figure 3.7 illustrates the centroid-owns-cell policy and ray clipping procedure.



(a) Centroid-owns-cell: only centroids inside the tile assign the cell

(b) Ray clipping

Figure 3.7. Annotation-aware raycast splitting. (a) The ROI is divided into overlapping tiles. Only cells whose centroids fall inside a given tile are assigned to it (centroid-owns-cell policy). (b) For each assigned cell, rays extending beyond the tile boundary are clipped to the nearest wall.

Padding.

After stain normalization, tiles with content smaller than $S \times S$ are padded with white pixels at the bottom and right edges. White padding is chosen because H&E slide backgrounds are white, so the model sees the same signal at tile edges as at slide borders. Bottom-right placement preserves annotation validity: the top-left origin $(0, 0)$ remains unchanged, so centroid positions and ray distances require no modification. The original content dimensions are recorded so that the training data loader can constrain random crop origins to the tissue-containing region.

3.2.1.3 Data Augmentation Strategies

3.2.1.3.1 RayCast Permutations.

Standard detection augmentations assume axis-aligned bounding boxes. Raycast annotations require specialized handling: spatial transforms (flips, rotations) permute the ray distance vector without changing magnitudes; scale multiplies all distances uniformly; and crops require ray clipping and a survival-rate filter to discard cells whose shape becomes unreliable.

All ray permutations depend only on R and the transform type, so they are precomputed at configuration time:

Table 3.2. Precomputed ray permutation indices for each geometric transform.

Transform	**Permutation**	**Example** ($R = 32$)
Horizontal flip	$\pi_{\text{flip-h}}(r) = (R/2 - r) \bmod R$	ray $0 \to 16$
Vertical flip	$\pi_{\text{flip-v}}(r) = (R - r) \bmod R$	ray $0 \to 0$, ray $1 \to 31$
Rotate 90° CCW	$\pi_{\text{rot}}(r; 1) = (r + R/4) \bmod R$	ray $0 \to 8$
Rotate 180°	$\pi_{\text{rot}}(r; 2) = (r + R/2) \bmod R$	ray $0 \to 16$
Rotate 270° CCW	$\pi_{\text{rot}}(r; 3) = (r + 3R/4) \bmod R$	ray $0 \to 24$

Any augmentation that changes the spatial relationship between annotations and the canvas triggers a four-step clipping pipeline: (1) discard cells whose centroid falls outside the crop region, (2) translate surviving centroids to crop-relative coordinates, (3) for each ray, compute the distance to the four canvas boundaries and clip the ray to $d'_r = \min(d_r, d_{\max}(r))$, and (4) discard cells with fewer than $\tau_{\text{surv}} = 0.3$ surviving rays. The boundary distance $d_{\max}(r)$ is computed identically to the chopping procedure described above, using the same ray-by-ray intersection with tile edges.

3.2.1.3.2 Augmentation Operations.

The following augmentations are applied during training:

Random crop. Crop origin is constrained to the tissue-containing region to avoid sampling white padding. Annotations whose centroid falls outside the crop are dropped; remaining cells are translated to crop-relative space and rays are clipped.

Stain jitter. A photometric-only augmentation in HSV space that mimics inter-slide stain variation without modifying any annotation field:

$$H' = H + \delta_H, \quad S' = S \cdot \gamma_S, \quad V' = V \cdot \gamma_V \tag{3-22}$$

where $\delta_H \sim \mathcal{U}[-9^\circ, 9^\circ]$, $\gamma_S \sim \mathcal{U}[0.7, 1.3]$, and $\gamma_V \sim \mathcal{U}[0.8, 1.2]$. An optional Gaussian blur is applied with probability 0.2 ($\sigma \sim \mathcal{U}[0.5, 1.0]$).

Random scale. Isotropic scaling by $s \sim \mathcal{U}[0.7, 1.3]$: all spatial quantities (centroid, all ray distances) are multiplied by s. For $s > 1$, a random crop restores the image to $S \times S$; for $s \leq 1$, random padding places the scaled image on a white canvas. Boundary clipping follows.

Random translation. Shift by $(\Delta x, \Delta y)$ with $|\Delta x|, |\Delta y| \leq \lfloor 0.1 \cdot S \rfloor$. Only

centroids move (ray distances are translation-invariant), followed by boundary clipping.

Horizontal flip. Applied with probability 0.5: centroid $c'_x = S - c_x$, and ray distances are permuted by the precomputed flip index.

3.2.2 Architecture Design

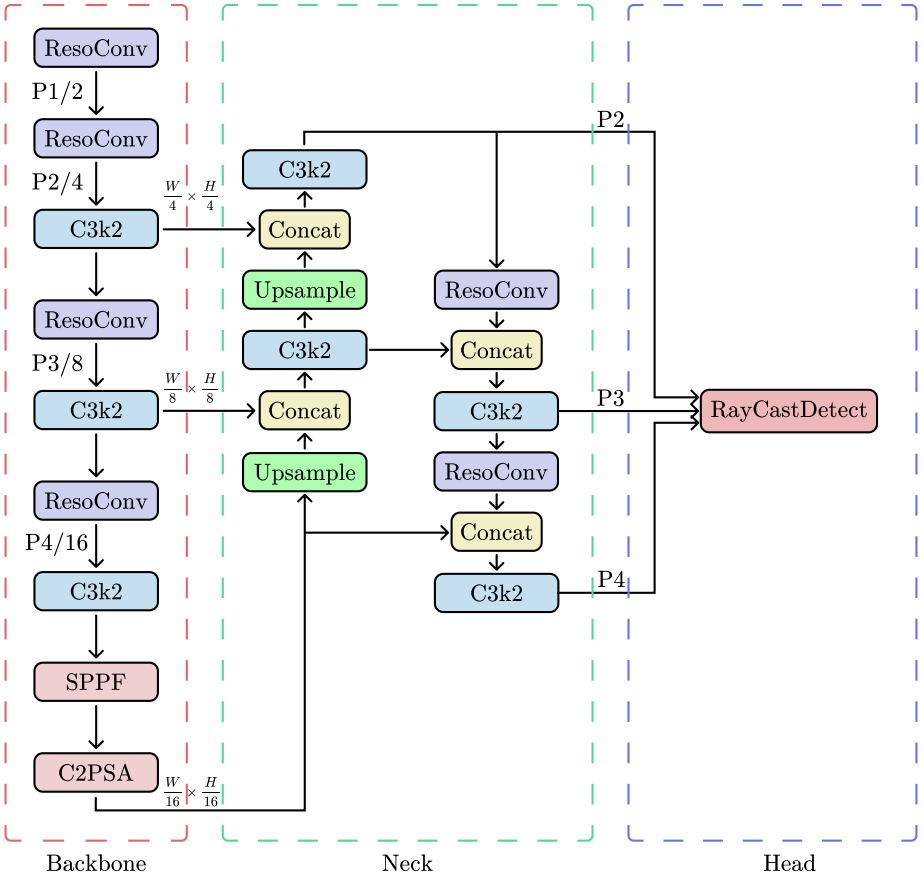


Figure 3.8. RayCastED model architecture

3.2.2.1 Design Considerations

Contour-based instance segmentation shares its architectural structure with object detection rather than with mask-based segmentation. Both contour-based instance segmentation and detection operate on an anchor grid, predict a sparse set of geometric parameters per location, and train through matching mechanisms rather than per-pixel classification. The star-convex ray representation used in this work replaces the bounding box regression target with polygon boundary distances. This is a change to the regression target, not to the underlying prediction paradigm, making the mature detection architecture ecosystem directly applicable. Four requirements constrain the base model choice: a fully convolutional backbone for lightweight inference, single-stage prediction for computational

efficiency, an established architecture for reproducibility, and NMS-free operation to avoid the density-dependent counting bias described in Section 2.2.3.2.4. YOLO26 satisfies all four through its dual-label assignment strategy 2.2.8.3, which enables one-to-one matching at inference without post-processing. Its implementation within the Ultralytics framework further provides optimized training pipelines and export pathways for edge deployment.

YOLO26 is designed for general objects and employs feature pyramid levels P3 through P5 (strides 8, 16, and 32). For a 256×256 input, these produce feature maps of 32×32, 16×16, and 8×8. Nuclei in H&E histopathology at 0.25 MPP typically measure 7 to 20 μm in diameter (28 to 80 pixels), making them small and densely packed relative to general objects. The P5 level at 8×8 resolution is too coarse to localize individual nuclei, and even P3 at 32×32 may miss the smallest cells. RayCastED therefore shifts the active pyramid levels to P2 through P4 (strides 4, 8, and 16), yielding a finest feature map of 64×64 for precise centroid localization. Extending to P2 through P5 is possible but adds anchor positions and backbone complexity that are redundant for nuclei-sized objects, which rarely require the receptive field of P5 2.2.3.2.3.

Standard downsampling operations (strided convolution and max pooling) act as low-pass filters that discard high-frequency content during feature map reduction. For large objects, this is acceptable because discriminative features reside in coarse shape and texture. For small nuclei, however, much of the discriminative signal lies in fine edges, chromatin texture, and boundary transitions that are smoothed away during standard downsampling 2.2.6.2. Discrete Wavelet Transform (DWT) based downsampling addresses this by decomposing the feature map into LL (approximation) and LH, HL, HH (detail) subbands. The LL subband serves as the downsampled representation while the detail subbands are preserved rather than discarded, retaining high-frequency information that would otherwise be lost 2.2.6.3.

The P2 through P4 configuration generates approximately 5,376 anchor locations for a 256×256 input ($64^2 + 32^2 + 16^2$). A typical histopathology tile contains on the order of 100 nuclei, yielding a foreground-to-background ratio of roughly 100/5,376 ($\approx$1.9%). A single classification head forced to simultaneously discriminate foreground from background and classify cell types under such extreme imbalance produces weak gradients for both tasks. RayCastED addresses this by splitting classification into a binary head (foreground/background, 1 channel) trained on all anchors and a multiclass head (cell type, n_c channels) trained only on foreground anchors. At inference, the final score is $\sigma(\text{binary}) \times \text{softmax}(\text{class})$, ensuring that only confident foreground predictions contribute to class probabilities.

Dual-label assignment 2.2.8.3 eliminates NMS by enforcing one-to-one matching within each pyramid level, but does not address cross-scale duplicates: the same nucleus may produce confident predictions at two adjacent pyramid levels simultaneously 2.2.3.2.3.

Conventional NMS is unsuitable for resolving these duplicates in dense nuclei, as valid detections of adjacent cells are indistinguishable from duplicate predictions based on overlap alone 2.2.3.2.4. Inter-scale competition complements dual-label assignment by normalizing classification scores across the scale dimension via softmax before output decoding. This operation suppresses lower-confidence cross-scale predictions during both training and inference in a fully differentiable manner, without requiring a hand-tuned IoU threshold.

3.2.2.2 Proposed Architecture Components

3.2.2.2.1 ResoConv.

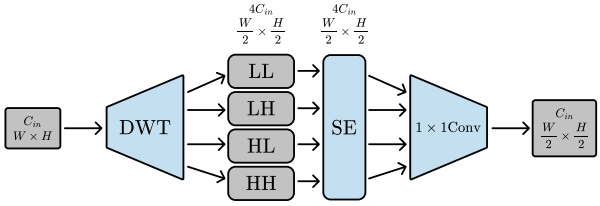


Figure 3.9. ResoConv

ResoConv replaces each strided downsampling step in the backbone and neck with a Haar DWT decomposition. The transform is implemented as a depthwise convolution with stride 2, where the four Haar filters are pre-computed and registered as non-trainable buffers. Depthwise convolution is used here for two reasons. First, the wavelet decomposition is inherently a per-channel operation: each input channel represents an independent feature map that should be decomposed independently, without cross-channel mixing during the transform itself. Cross-channel combination is deferred to the subsequent 1×1 projection. Second, depthwise convolution reduces the computational cost from $K^2 \cdot C^2$ to $K^2 \cdot C$ per spatial position 2.2.4.3, which is significant given that ResoConv replaces every downsampling layer in both the backbone and neck. Unlike standard strided convolution, which acts as a learned low-pass filter and discards high-frequency content irreversibly, the DWT preserves this information explicitly across the four subbands 2.2.6.3.

The four subbands are concatenated along the channel dimension and passed through a Squeeze-and-Excitation (SE) channel attention module. SE performs global average pooling followed by a bottleneck fully-connected layer, a ReLU activation, a restoring layer, and a sigmoid gate that produces per-channel weights. This allows the network to learn the relative importance of each subband at each layer: in early layers, high-frequency detail subbands may receive higher weights for boundary delineation, while in deeper layers, the LL approximation may dominate for semantic feature extraction.

The SE-reweighted features are then projected to the target channel count via a 1×1 convolution, producing the final output at half spatial resolution.

3.2.2.2.2 RayCastDetect.

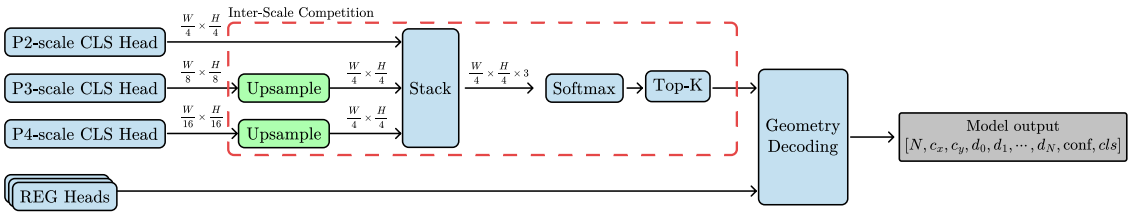


Figure 3.10. RayCastDetect

RayCastDetect extends the YOLO detection head to predict star-convex polygon parameters instead of axis-aligned bounding boxes. For each anchor, the regression branch produces $2 + R$ values: two centroid offsets and $R = 64$ ray distances radiating from the centroid at evenly spaced angles. The classification branch is split into a binary foreground/background head and a multiclass head with n_c channels, yielding a total of $3 + R + n_c$ output values per anchor. The head is trained with dual-label assignment 2.2.8.3 using a one-to-many (o2m) head for rich supervision and a one-to-one (o2o) head for NMS-free inference; the o2m head is discarded after training. Three key design choices distinguish RayCastDetect from the standard YOLO head: inter-scale competition to suppress cross-scale duplicate detections, a split classification head to address foreground-background imbalance, and a regression head with RayRefinementBlock to smooth spatially inconsistent ray predictions.

Inter-scale Competition. Operating across three feature scales (P2, P3, P4), the head often assigns high confidence to the same nucleus at multiple scales simultaneously, producing cross-scale duplicate detections. Inter-scale competition addresses this by upsampling the classification scores from all scales to P2 resolution, applying softmax across the scale dimension, and multiplying each scale's confidence by its resulting competition weight. This mechanism is fully differentiable and operates during both training and inference, providing a learned, soft suppression that complements (rather than replaces) the dual-label assignment strategy 2.2.8.3. Unlike NMS 2.2.3.2.4, which requires a hand-tuned IoU threshold and is applied post-hoc, inter-scale competition is integrated into the network and suppresses duplicates through gradient-descent-optimized weights.

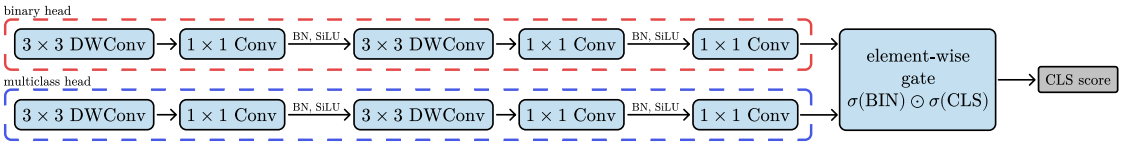


Figure 3.11. Split Classification Head

Split Classification Head. The P2-P4 scale configuration at 256×256 pixel input produces approximately 5,376 anchors per image, yet a typical histopathology image contains fewer than 100 nuclei. This extreme foreground-background imbalance makes joint objectness and classification difficult for a single head. RayCastDetect therefore splits the classification branch into a binary head with one channel that is applied to all anchors, and a multiclass head with n_c channels that is supervised only on foreground anchors. At inference, the two are combined as $P(\text{fg}) \times P(\text{type} \mid \text{fg})$, i.e., $\text{sigmoid}(\text{binary}) \times \text{softmax}(\text{class})$. This decomposition separates the easier binary problem of distinguishing nucleus from background, where the binary head can exploit all anchor signals, from the harder multiclass problem of cell-type discrimination, where the class head focuses exclusively on foreground samples.

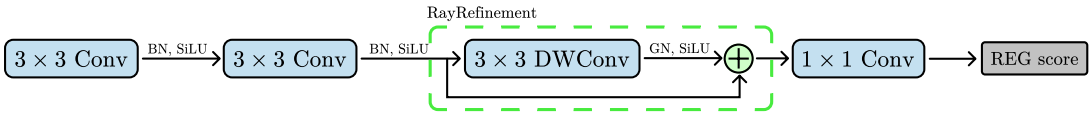


Figure 3.12. Regression Head with RayCast Refinement

Regression Head. The regression branch produces the polygon parameters per anchor through two 3×3 convolution layers, a RayRefinementBlock, and a final 1×1 convolution that outputs $2 + R = 66$ channels: two centroid offsets and 64 ray distances. The RayRefinementBlock applies a 3×3 depthwise convolution followed by GroupNorm and SiLU activation with a residual skip connection. GroupNorm is preferred over BatchNorm because each scale level (P2, P3, P4) maintains independent head weights, including its own RayRefinementBlock, and the three scales operate at different spatial resolutions (64×64, 32×32, 16×16) with different object distributions: P2 densely covers small nuclei, while P4 sparsely covers larger structures. BatchNorm would maintain separate running mean and variance statistics per scale computed over these inconsistent conditions, leading to unstable normalization when the statistics do not transfer cleanly. GroupNorm normalizes within channel groups per sample, avoiding any cross-scale or batch-dimension dependency. The residual connection ensures the block refines rather

than replaces the existing features: since the purpose is to smooth spatially inconsistent ray predictions by blending neighboring anchor features, the skip connection guarantees that the base prediction is preserved when no correction is needed. The two centroid channels use sigmoid activation to produce bounded $[0, 1]$ grid-cell offsets, while the 64 ray channels use softplus activation to ensure strictly positive distances. Bias initialization sets the ray channels to the minimum plausible nucleus size using the formula $(d_{\min} + \epsilon)/(2 \cdot S \cdot \rho)$, where $d_{\min} = 3.0$ μm is the minimum nucleus diameter, $\epsilon = 0.5$ μm is a linear approximation buffer, $S = 256$ is the crop size in pixels, and $\rho = 0.25$ μm/pixel is the scanning resolution. This yields a normalized target of $3.5/128 \approx 0.027$, corresponding to a 7 pixel radius at 0.25 μm/pixel. The inverse softplus of this value is used as the bias so that softplus(bias) produces the target directly at initialization, providing a unidirectional expansion gradient at the start of training.

3.2.3 Training Strategy and Optimization

3.2.3.1 Curriculum

Training uses the MuSGD optimizer, which applies the Muon optimizer 2.2.5.2 to two-dimensional weight matrices (convolution kernels, linear layers) and standard SGD with Nesterov momentum to one-dimensional parameters (biases, normalization scales). This hybrid leverages Muon's orthogonalized gradient updates for the high-rank weight matrices that dominate model capacity while retaining SGD stability for scalar parameters. The base learning rate is 0.01, momentum 0.937, weight decay 5×10^{-4}, and gradients are clipped to a maximum norm of 10.0. Training uses a batch size of 16 on 256×256 pixel crops.

The learning rate follows cosine annealing with warm restarts (SGDR) [40], with initial cycle length $T_0 = 133$ epochs, cycle multiplier $T_{\text{mult}} = 1$, and minimum learning rate $\eta_{\min} = 10^{-4}$. This produces three equal cycles over the full training run. Each restart abruptly increases the learning rate, helping the optimizer escape sharp minima and explore broader regions of the loss landscape before annealing again.

Three annealing schedules progressively shift the training objective. First, the o2m and o2o loss weights transition linearly from $0.8/0.2$ to $0.1/0.9$, gradually shifting from dense one-to-many supervision to clean one-to-one matching for NMS-free inference 2.2.8.3. Second, the o2o assigner's topk2 parameter anneals from 3 to 1 (one-to-few to one-to-one), beginning at epoch 67 and completing at 50% of the training schedule, allowing multiple positive anchors per ground truth during early feature learning before converging to the strict bipartite matching required at inference. Third, the smoothness loss weight λ_{smooth} ramps from 0 to 3 over the first 40% of epochs, letting the ray predictions fit target shapes before enforcing boundary regularization.

3.2.3.2 Assignment

RayCastDetect uses two parallel instances of RayCastAssigner, which extends the task-aligned matching 2.2.8.2 strategy by replacing standard IoU with PolarIoU in the alignment metric. The one-to-many branch uses topk $= 15$ and $\text{topk}_2 = 15$ for dense multi-anchor supervision, while the one-to-one branch uses topk $= 7$ (candidate pool) and $\text{topk}_2 = 3 \to 1$ (annealed). The o2m alignment metric is:

$$t = s^{\alpha} \cdot \text{PolarIoU}^{\beta} \cdot \exp\left(\frac{-d^2}{2\sigma^2}\right) \tag{3-23}$$

where s is the classification score, $\alpha = 0.5$, $\beta = 2.0$, d is the centroid-to-anchor distance, and σ is the per-nucleus radius computed as the 75th-percentile non-zero ray distance multiplied by a scale factor of 1.0. The inclusion of PolarIoU in the o2m metric ensures that early training prioritizes geometric quality: anchors are rewarded for producing well-shaped polygons, not just confident classifications. The Gaussian spatial decay term $\exp(-d^2/2\sigma^2)$ restricts high alignment scores to spatially proximate anchors, with σ adapted to each nucleus size.

The o2o branch uses classification-only task-aligned matching ($\text{cls}^{\alpha} \cdot \exp(-d^2/2\sigma^2)$, without PolarIoU), forcing the classification head to become the primary ranking signal. This matches inference conditions, where no localization feedback is available to resolve assignment ambiguity, and creates a self-reinforcing dynamic: higher classification confidence yields better assignments, which in turn produce stronger classification gradients.

3.2.3.3 Loss Function

The total training loss combines five terms:

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{xy}}\mathcal{L}_{\text{xy}} + \lambda_{\text{L1}}\mathcal{L}_{\text{L1}} + \lambda_{\text{piou}}\mathcal{L}_{\text{piou}} + \lambda_{\text{smooth}}\mathcal{L}_{\text{smooth}} \tag{3-24}$$

with weights $\lambda_{\text{cls}} = 2.0$, $\lambda_{\text{xy}} = 500.0$, $\lambda_{\text{L1}} = 14.0$, $\lambda_{\text{piou}} = 3.0$, and λ_{smooth} annealed from 0 to 3 over the first 40% of training. Each branch (o2m and o2o) computes this loss independently, and the two are combined using the o2m/o2o weight schedule described above.

The classification loss differs between branches. The o2m head uses focal loss ($\gamma = 2.0$, $\alpha = 0.75$) with background subsampling at a 3:1 ratio of background to foreground anchors. The o2o head, using the split classification design, applies focal loss ($\gamma = 2.0$, $\alpha = 0.75$) on the binary foreground/background head across all anchors, and binary cross-entropy on the multiclass head restricted to foreground anchors only.

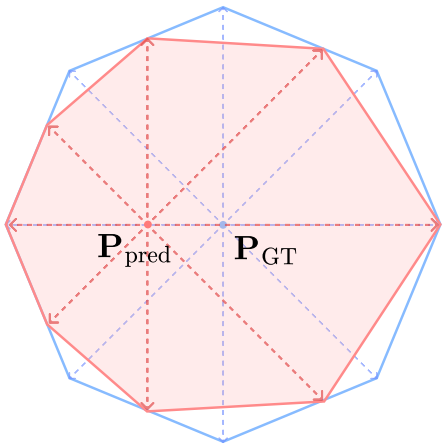


Figure 3.13. Ray distance loss computation. The analytical ray distances are recomputed from the predicted centroid, and the log-space L1 loss is evaluated between the predicted and analytical distances.

The ray distance loss $\mathcal{L}_{\text{L1}}$ is the most distinctive component. During data ingestion, ground-truth ray distances are computed by casting rays from the annotation centroid to the polygon boundary using geometric line intersection. During training, this same raycasting operation is reformulated as a differentiable Cramer's rule solver that computes analytical ray distances from the *predicted* centroid (with gradients detached) to the ground-truth polygon boundary. The loss is then evaluated in log-space:

$$\mathcal{L}_{\text{L1}} = \frac{1}{R} \sum_{i=1}^{R} \left| \log(\hat{d}_i + \varepsilon) - \log(d_i^{\text{analytical}} + \varepsilon) \right| \tag{3-25}$$

This analytical recomputation couples centroid accuracy from ray supervision: the ray head is always judged against geometrically correct targets for the position where the model actually placed the centroid, rather than against static targets computed from the ground-truth centroid. If the predicted centroid shifts from the ground-truth centroid, the target rays adapt accordingly, preventing conflicting gradient signals between the centroid and ray branches. Log-space evaluation provides scale-invariant gradients, applying equal relative penalty for over-prediction and under-prediction regardless of absolute ray length.

The centroid loss $\mathcal{L}_{\text{xy}}$ uses Huber loss ($\delta = 0.05$) on normalized grid-cell offsets. The high weight $\lambda_{\text{xy}} = 500$ compensates for gradient attenuation caused by the sigmoid activation, stride multiplication, and pixel-space normalization that the offsets undergo during decoding.

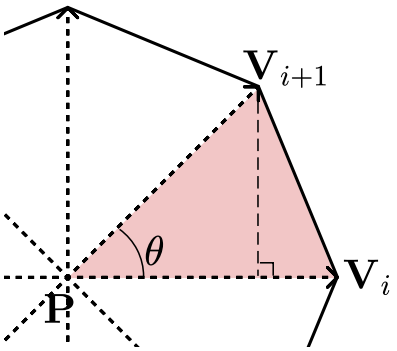


Figure 3.14. PolarIoU loss computed through sector-area decomposition. Each adjacent pair of rays forms a triangular sector, and the IoU is computed analytically from the sector intersections without rasterization.

The PolarIoU loss provides shape-aware geometric supervision. PolarMask [7] originally proposed a PolarIoU based on per-ray distance comparison:

$$\text{PolarIoU}_{\text{PolarMask}} = \frac{\sum_{i=0}^{R-1} \min(d_i, \hat{d}_i)}{\sum_{i=0}^{R-1} \max(d_i, \hat{d}_i)} \tag{3-26}$$

While straightforward, this formulation treats each ray independently and does not capture the geometric interaction between adjacent rays.

We modify PolarIoU to be sector-area based, where each sector is a triangular region formed by two adjacent rays i and $i+1$ and the centroid. The area of a sector with included angle $\Delta\theta = 2\pi/R$ is:

$$A_i = \frac{1}{2} \cdot d_i \cdot d_{i+1} \cdot \sin\left(\frac{2\pi}{R}\right) \tag{3-27}$$

Since $\frac{1}{2} \cdot \sin(2\pi/R)$ is identical for all sectors, the IoU ratio cancels this constant factor, yielding:

$$\text{PolarIoU} = \frac{\sum_{i=0}^{R-1} \min(d_i, \hat{d}_i) \cdot \min(d_{i+1}, \hat{d}_{i+1})}{\sum_{i=0}^{R-1} \max(d_i, \hat{d}_i) \cdot \max(d_{i+1}, \hat{d}_{i+1})} \tag{3-28}$$

with cyclic indexing $d_R \equiv d_0$, and the loss is $\mathcal{L}_{\text{piou}} = -\log(\text{PolarIoU})$. The sector-area formulation couples adjacent rays through the product $d_i \cdot d_{i+1}$, ensuring each ray prediction contributes to two overlapping sectors and propagating gradient information

across angular neighbors. Additionally, our formulation computes PolarIoU using the analytical ground-truth rays derived from the predicted centroid (consistent with $\mathcal{L}_{\text{L1}}$), rather than static ground-truth rays as in the original PolarMask.

The smoothness loss applies a 1D discrete Laplacian on the circular sequence of ray distances. Treating the R ray distances as a signal $\{d_i\}_{i=0}^{R-1}$ on a periodic domain, the filter kernel $[-1,\ 2,\ -1]$ measures deviation from local linearity:

$$\mathcal{L}_{\text{smooth}} = \frac{1}{R}\sum_{i=0}^{R-1} |d_{i+1} - 2d_i + d_{i-1}| \tag{3-29}$$

with cyclic indexing $d_{-1} \equiv d_{R-1}$ and $d_R \equiv d_0$. When all $d_i = \rho$ (a constant), the Laplacian is zero everywhere; a constant-distance sequence reconstructs to a perfect circle in pixel space. The L1 norm is chosen over L2 for sparsity: along sections where the boundary evolves linearly, the Laplacian vanishes exactly, while sharp corners at specific orientations incur a bounded penalty. L2 would diffuse the penalty across all indices, rounding every transition regardless of biological validity. This loss does not push predictions toward any particular shape; it only penalizes non-smooth variation along the ray sequence, allowing elongated and biologically plausible forms as long as the ray distances change linearly. The weight λ_{smooth} is annealed from 0 to 3 over the first 40% of training.

3.2.4 Evaluation Procedure

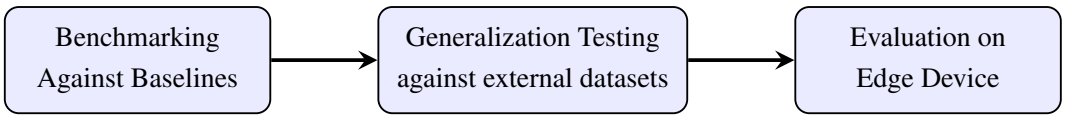


Figure 3.15. Evaluation Procedure Diagram

3.2.4.1 Evaluation Metrics

Predicted star-convex polygons are rasterized to binary masks for overlap-based evaluation. When multiple predicted masks overlap at the same pixel, the largest mask receives priority. All PQ-family metrics use Hungarian matching between predicted and ground-truth masks at IoU ≥ 0.5. The F1 score, precision, and recall use centroid-based matching at a distance threshold of 12 pixels, and AJI uses its own global intersection-over-union formulation 2.2.9.4. During training validation, bPQ is computed from PolarIoU (analytical, no rasterization required) as a fast proxy; final evaluation uses rasterized mask IoU for accuracy.

Three categories of metrics are reported. Detection quality is measured by F1, precision, and recall. Segmentation quality is measured by bPQ (binary, all classes

merged), mPQ (per-class, averaged over the five PanNuke nuclei types), and AJI (global instance-level). Efficiency is measured by GFLOPs, parameter count, and runtime in milliseconds per image. The model selection fitness during training is $0.7 \times \text{bPQ} + 0.3 \times \text{mAP}_{50}$, following the Ultralytics convention where mAP at IoU 0.5 provides detection-quality feedback alongside the segmentation-focused bPQ.

3.2.4.2 Benchmarking

RayCastED is trained on PanNuke Fold 1, validated on Fold 2, and tested on Fold 3. The Base (B) variant is compared against StarDist [8], LKCell [16], HoVer-NeXt [14], and CellPose-SAM. All baselines are trained and evaluated on the same PanNuke fold split under their respective published configurations. A confidence threshold of 0.20 is used for all RayCastED predictions to test the model's raw ranking quality rather than relying on aggressive thresholding to suppress false positives. All metrics described above are reported in a comprehensive comparison table.

3.2.4.3 Generalization Test

To assess cross-dataset generalization, RayCastED trained on PanNuke Fold 1 is evaluated without fine-tuning on MoNuSAC [38] and PUMA. Only AJI, bPQ, and F1 are reported to avoid dataset taxonomy clashes: MoNuSAC defines four nuclei classes and PUMA uses a different class schema, making per-class mPQ comparisons across datasets misleading. Class labels from target datasets are harmonized to PanNuke's taxonomy where unambiguous mappings exist; where mappings are unclear, evaluation collapses to binary foreground/background. Results indicate whether features learned across PanNuke's 19 tissue types transfer to unseen tissues, staining protocols, and scanning resolutions.

3.2.4.4 Evaluation on Edge Device

Two runtime backends are benchmarked on the desktop GPU (RTX 5070 Ti): native PyTorch and TensorRT. The PyTorch runtime serves as the accuracy reference and the TensorRT runtime establishes the performance ceiling achievable on a high-power GPU before edge deployment.

For edge deployment, the model is exported to ONNX (opset 17, graph simplified) and deployed on a Jetson Orin Nano. The ONNX graph exports raw logits only; all decoding (sigmoid for centroid offsets, softplus for ray distances, grid decoding, hierarchical classification combination) and postprocessing (top-k selection, confidence filtering, polygon scaling) are performed in a Python layer on-device. Three edge runtime configurations are evaluated: ONNX Runtime, TensorRT FP32, and TensorRT FP16. All configurations are tested on PanNuke Fold 3 for accuracy parity with the desktop results.

Runtime is measured via wall-clock timing over all test images, reported as milliseconds per image. All metrics except GFLOPs and parameter count are reported, since the architecture is identical across backends. The FP32 versus FP16 comparison quantifies the accuracy-speed trade-off of half-precision inference on edge hardware.

3.3 Research Workflow

The development of RayCastED followed four sequential phases, each building on the output of the preceding one. Figure 3.16 illustrates this workflow.

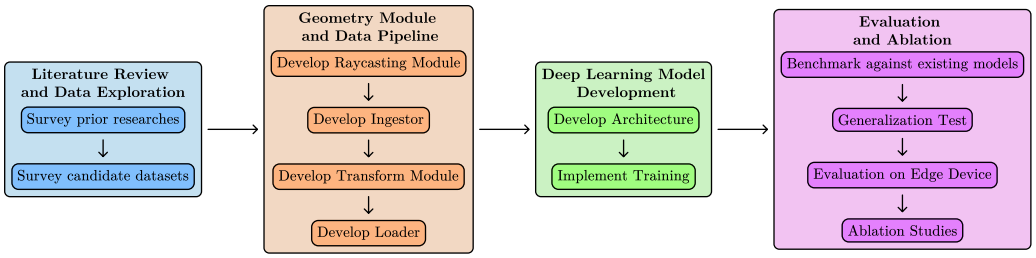


Figure 3.16. Research workflow organized into four sequential phases: literature review and data exploration, geometry module and data pipeline development, deep learning model development, and evaluation and ablation.

Phase 1: Literature Review and Data Exploration.

The research began with a survey of existing nuclei detection and segmentation approaches, including contour-based methods (StarDist [8]), cell border prediction (CellPose [17]), and multi-task learning architectures (LKCell [16], HoVer-NeXt [14]). Candidate datasets were surveyed for annotation format, tissue coverage, staining protocol, and compatibility with the raycast representation.

Phase 2: Geometry Module and Data Pipeline Development.

The raycasting module was developed to compute ray-edge intersections. A data pre-processing pipeline was then built, consisting of three stages: data ingestion to read diverse annotation formats, data transformation including stain normalization and tiling, and a data loader with raycast-aware augmentation.

Phase 3: Deep Learning Model Development.

A base object detection architecture was selected based on four criteria: a fully convolutional backbone for lightweight inference, single-stage prediction for computational efficiency, an established codebase for reproducibility, and NMS-free operation.

The architecture was then modified for nuclei instance segmentation through a custom downsampling method and a task-specific prediction head. The training pipeline was implemented with a cosine annealing learning rate schedule and a combined loss function suited to the contour-based regression objective.

Phase 4: Evaluation and Ablation.

RayCastED was benchmarked against existing methods using detection, segmentation, and efficiency metrics. Zero-shot generalization was tested on unseen datasets. Edge deployment feasibility was validated on the Jetson Orin Nano. Ablation studies were conducted on the architectural components and training configurations.

CHAPTER IV
RESULTS AND DISCUSSION

This chapter presents the experimental evaluation of RayCastED across the research objectives outlined in Chapter 1. The experiments are designed to assess both detection accuracy and computational efficiency, covering the RayCast representation fidelity, full benchmarking against state-of-the-art methods, cross-dataset generalization, and edge deployment feasibility.

4.1 Performance Evaluation

We first validate the RayCast representation itself, then benchmark RayCastED against prior methods, and finally test generalization to unseen datasets.

4.1.1 RayCast Representation Validation

To assess how faithfully the ray-cast polygon approximates the ground truth mask, we compute the mask IoU between the original segmentation and the polygon reconstructed from R equally-spaced rays. Table 4.1 reports the per-tissue IoU for $R \in \{8, 16, 32, 64\}$ across the five organ types in PanNuke.

Table 4.1. Summary statistics of the mask IoU between ground truth nuclei and the ray-cast polygon approximation at varying numbers of rays R (mean, standard deviation, median, minimum, and maximum IoU computed over 1,849 nuclei from the PanNuke validation set).

Statistic	$R = 8$	$R = 16$	$R = 32$	$R = 64$
Mean	0.8099	0.8810	0.9009	0.9078
STD	0.0957	0.0901	0.0795	0.0775
Median	0.8425	0.9060	0.9195	0.9246
Min	0.2727	0.2727	0.2667	0.2667
Max	0.9224	0.9764	0.9843	0.9902

As expected, increasing R produces tighter approximations. The gain from $R = 16$ to $R = 32$ (Mean IoU $+0.0199$, Max $+0.0205$) is substantial, while the further increase to $R = 64$ yields only incremental improvement (Mean $+0.0069$, Max $+0.0034$), indicating diminishing returns beyond 32 rays. Although $R = 32$ offers the best efficiency–accuracy trade-off, $R = 64$ is used as the default in all subsequent experiments to maximize boundary fidelity without a significant computational penalty.

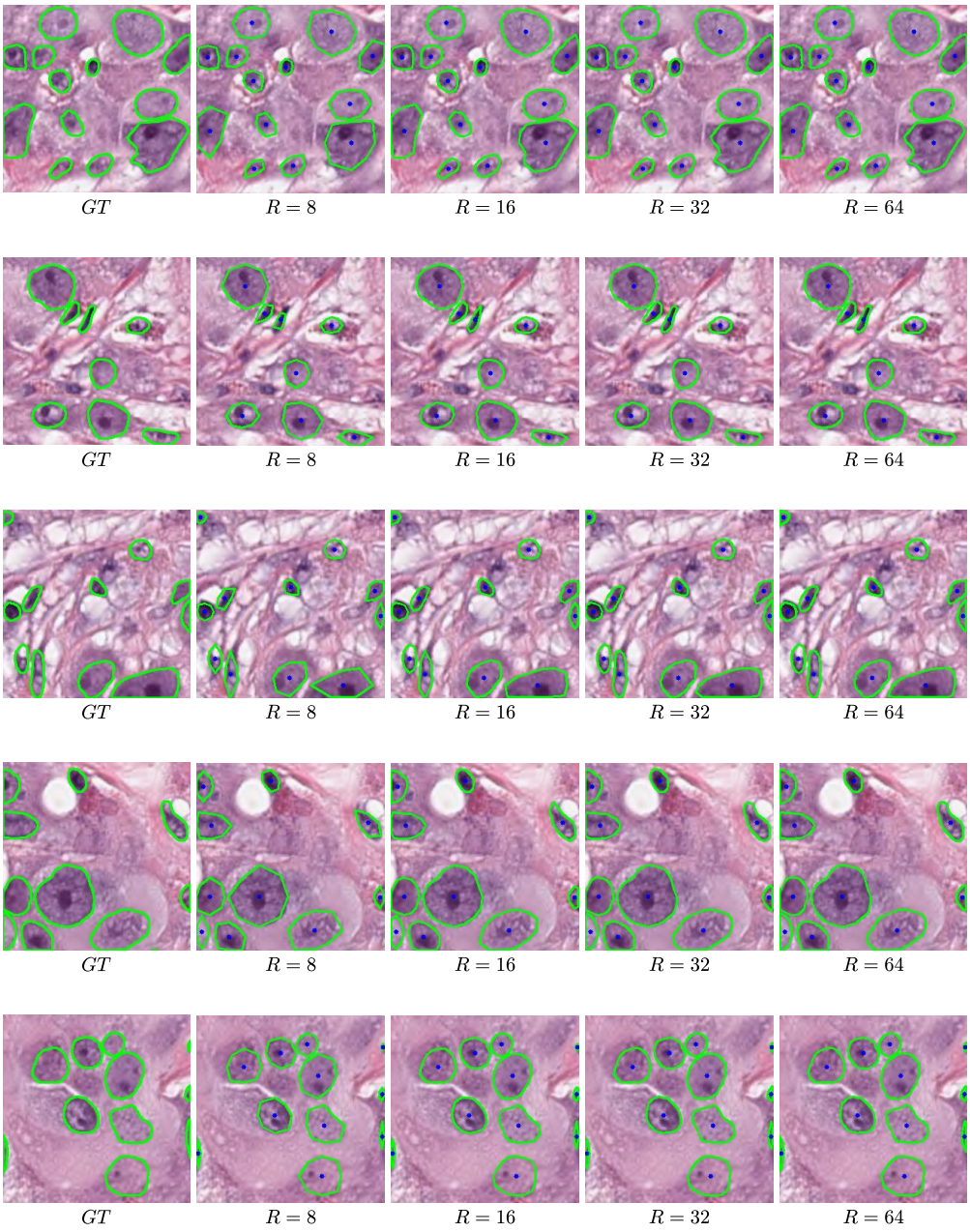


Figure 4.17. Visual comparison of ground truth nucleus mask and ray-cast polygon reconstruction at $R = 8, 16, 32, 64$.

4.1.2 RayCastED Benchmarking Results

We compare RayCastED against four prior methods on the PanNuke test set. All models are evaluated under the same protocol on a held-out set of 4,716 nuclei across 79 images. Table 4.2 reports detection and segmentation accuracy, while Table 4.3 reports model complexity and inference speed.

Table 4.2. Accuracy benchmarking results on the PanNuke test set. Best overall is shown in **bold**.

Method	AJI	bPQ	mPQ	F1	Precision	Recall
StarDist	0.5715	0.5945	0.3670	0.7806	0.8558	0.7176
CellPose-SAM	**0.6314**	0.6028	–	0.7969	**0.8687**	0.7360
LKCell	0.6228	**0.6793**	**0.5202**	**0.8147**	0.8216	**0.8079**
HoVer-NeXt	0.5680	0.6521	0.4708	0.7899	0.8320	0.7520
RayCastED-S	0.5555	0.5780	0.4337	0.7176	0.7146	0.7206
RayCastED-B	0.5715	0.6024	0.4526	0.7409	0.7419	0.7400
RayCastED-L	0.5740	0.5947	0.4764	0.7591	0.8113	0.7132

CellPose-SAM and LKCell dominate the accuracy rankings, which is expected given their model complexity: as shown in Table 4.3, both operate at parameter counts and GFLOPs well above the RayCastED family, with super-linear time complexity in LKCell. RayCastED nonetheless delivers competitive performance. In particular, RayCastED-L achieves metrics within 10% of the best result in every column while requiring a fraction of the parameters and GFLOPs and scaling with linear time complexity.

Table 4.3. Model complexity and time complexity benchmarking. Best overall is shown in **bold**.

Method	Params	GFLOPs	Time Complexity
StarDist	1.46M	8.14	$O(n + k^2)$
CellPose-SAM	$\sim$305M	724	$O(n^2)$
LKCell	163.8M	47.86	$O(n \log n + kn)$
HoVer-NeXt	36M	24	$O(n \log n + kn)$
RayCastED-S	**0.81M**	**1.38**	$O(n)$
RayCastED-B	2.6M	5.52	$O(n)$
RayCastED-L	9.44M	12.42	$O(n)$

RayCastED achieves linear time complexity as a direct consequence of its post-processing-free design: without NMS or watershed, inference scales proportionally to the

number of anchor positions rather than the number of detected objects. Since the anchor grid is fixed by input resolution and stride, this complexity is image-size agnostic – the model accepts arbitrarily sized inputs without a super-linear increase in cost, enabling throughput-consistent deployment across varying tile dimensions.

To provide deeper insight, we also break down the results by tissue type (Table 4.4) and by nuclei type (Table 4.5).

Table 4.4. Per-tissue bPQ and mPQ for all evaluated methods on the PanNuke test set across 19 organ types. The best value in each row is shown in **bold**.

	StarDist		CellPose-SAM		LKCell		HoVer-NeXt		RayCastED-S		RayCastED-B		RayCastED-L	
Tissue	**bPQ**	**mPQ**	**bPQ**	**mPQ**	**bPQ**	**mPQ**	**bPQ**	**mPQ**	**bPQ**	**mPQ**	**bPQ**	**mPQ**	**bPQ**	**mPQ**
Adrenal Gland	0.6046	0.4268	0.6476	–	**0.7221**	**0.6136**	0.7015	0.5731	0.6057	0.5259	0.6485	0.5545	0.6331	0.5794
Bile Duct	0.5750	0.4699	0.6146	–	**0.6666**	**0.5287**	0.6452	0.4974	0.5839	0.4438	0.6090	0.4755	0.6002	0.4859
Bladder	0.5631	0.5451	0.6180	–	**0.6754**	**0.6073**	0.6589	0.5021	0.5269	0.5005	0.6115	0.5427	0.6074	0.5524
Breast	0.6592	0.4595	0.6391	–	**0.7118**	**0.5880**	0.6846	0.5264	0.6206	0.4972	0.6346	0.5101	0.6290	0.5250
Cervix	0.5686	0.4122	0.5475	–	**0.6292**	**0.6058**	0.6095	0.5458	0.5278	0.4625	0.5327	0.4979	0.5545	0.5531
Colon	0.4599	0.4124	0.4855	–	**0.5707**	**0.4631**	0.5312	0.4330	0.4472	0.3993	0.4793	0.4160	0.4760	0.4434
Esophagus	0.6564	0.4420	0.6495	–	**0.7529**	**0.5270**	0.7105	0.5048	0.6651	0.4668	0.6874	0.4514	0.6692	0.4842
Head & Neck	0.5086	0.4482	0.5015	–	**0.5741**	**0.5424**	0.5411	0.4411	0.4931	0.4134	0.5243	0.4530	0.5185	0.4724
Kidney	0.5885	0.3574	0.5905	–	**0.6877**	0.5372	0.6591	**0.5384**	0.5929	0.3913	0.6035	0.4225	0.5817	0.4729
Liver	0.6740	0.4187	0.6602	–	**0.7801**	**0.6100**	0.7409	0.5527	0.6866	0.5102	0.7110	0.5090	0.6816	0.5368
Lung	0.5664	0.2871	0.5797	–	**0.7179**	**0.4315**	0.6962	0.4076	0.6159	0.3713	0.6677	0.3808	0.6365	0.4036
Ovarian	0.6669	0.4492	0.6681	–	**0.7446**	**0.6246**	0.7175	0.5144	0.6644	0.5189	0.6842	0.5376	0.6592	0.5674
Pancreatic	0.6441	0.4333	0.6428	–	**0.7413**	0.4895	0.7317	**0.5524**	0.6804	0.4530	0.6740	0.4661	0.6872	0.4692
Prostate	0.6215	0.4280	0.6416	–	**0.6927**	**0.6015**	0.6867	0.5361	0.6132	0.5091	0.6363	0.5021	0.6482	0.5646
Skin	0.4736	0.3451	0.5509	–	0.6115	0.4685	**0.6559**	**0.4736**	0.4576	0.4663	0.5281	0.4326	0.5374	0.4514
Stomach	0.7053	0.5222	0.7234	–	0.7564	**0.5582**	**0.7715**	0.4492	0.6837	0.3937	0.7216	0.4304	0.7005	0.4606
Testis	0.6302	0.4257	0.6470	–	**0.7124**	**0.5936**	0.6735	0.5562	0.6157	0.5005	0.6276	0.5257	0.6366	0.5466
Thyroid	0.6492	0.4232	0.6637	–	**0.7455**	**0.5754**	0.6983	0.5300	0.6529	0.4799	0.6487	0.4710	0.6472	0.5064
Uterus	0.6396	0.4511	0.6771	–	**0.7324**	0.4875	0.7149	0.3663	0.5949	0.4026	0.6226	0.3943	0.5986	**0.4892**
AVG	0.6029	0.4293	0.6183	–	**0.6961**	**0.5502**	0.6752	0.5000	0.5962	0.4582	0.6238	0.4723	0.6159	0.5034
STD	0.0679	0.0574	0.0615	–	**0.0612**	0.0593	0.0615	0.0570	0.0741	0.0496	0.0666	0.0513	0.0603	**0.0494**

The per-tissue breakdown reveals a nuanced picture beyond the global metrics. LKCell remains the strongest model across most organ types, with HoVer-NeXt occasionally taking the lead on Skin, Stomach, Kidney, and Pancreatic. Notably, RayCastED-L achieves the best mPQ on Uterus (0.4892) despite having only 10.3M parameters. More importantly, RayCastED-L exhibits the lowest standard deviation in per-tissue mPQ (0.0494), indicating the most consistent panoptic quality across heterogeneous organ types – a critical property for deployment on multi-organ pathology workflows where unpredictable tissue variation is the norm.

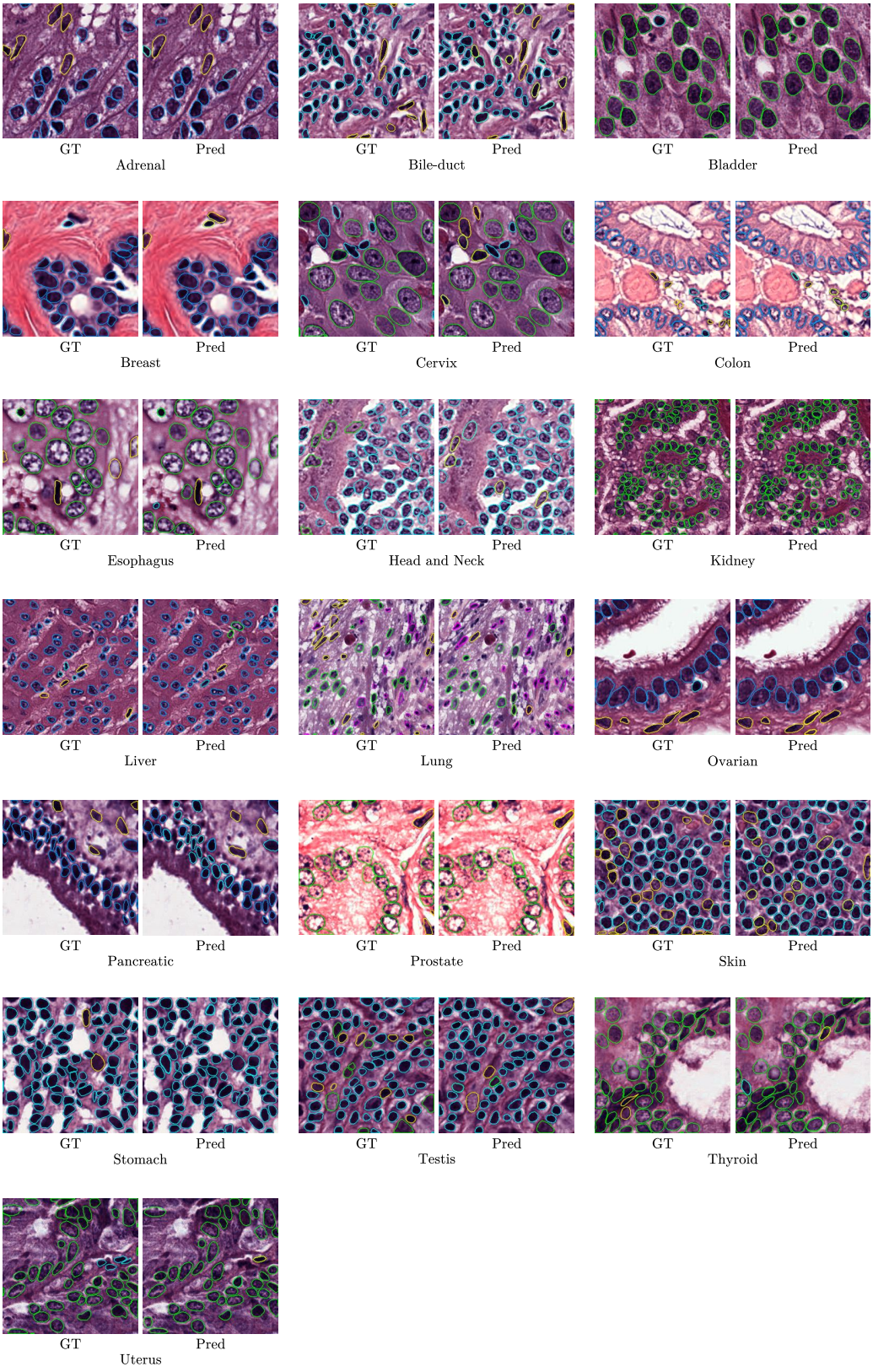


Figure 4.18. RayCastED-B predictions across the 19 PanNuke tissue types.

Table 4.5. Per-nuclei-type precision, recall, and F1-score on the PanNuke test set. Nuclei types follow the PanNuke five-class taxonomy. CellPose-SAM does not provide nuclei classification. The best value in each column is shown in **bold**.

	Neoplastic			Inflammatory			Connective			Necrosis			Epithelial		
Method	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
StarDist	0.6213	0.5998	0.6104	0.5857	0.5431	0.5636	0.5719	0.4009	0.4714	0.3333	0.0009	0.0019	0.4326	0.2998	0.3541
CellPose-SAM	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
LKCell	0.7285	**0.7167**	**0.7225**	**0.7090**	**0.6603**	**0.6838**	0.6067	**0.6002**	**0.6034**	0.4497	0.3340	0.3833	0.7019	**0.7459**	0.7232
HoVer-NeXt	**0.7372**	0.6566	0.6946	0.6932	0.6392	0.6651	**0.6207**	0.5722	0.5955	**0.4750**	0.3595	**0.4093**	**0.7630**	0.6909	**0.7252**
RayCastED-S	0.6775	0.6441	0.6604	0.5546	0.6009	0.5768	0.5105	0.5053	0.5079	0.3826	**0.4062**	0.3940	0.6000	0.6628	0.6298
RayCastED-B	0.6825	0.6718	0.6771	0.5751	0.5947	0.5848	0.5318	0.5322	0.5320	0.3919	0.3927	0.3923	0.6615	0.6534	0.6574
RayCastED-L	0.7329	0.6558	0.6922	0.6609	0.5907	0.6239	0.6029	0.5086	0.5518	0.4470	0.3532	0.3946	0.7339	0.6540	0.6916

The per-type analysis follows a similar pattern: LKCell dominates across most nuclear categories, with HoVer-NeXt claiming occasional wins on Epithelial and Necrosis F1. A surprising result emerges in the Necrosis category, where the RayCastED family consistently achieves the highest recall despite having an order of magnitude fewer parameters. This suggests that the split classification head and the hierarchical multi-objective loss provide an advantage for detecting rare minority classes that are often under-represented in the training data.

4.1.3 Generalization Test

To evaluate cross-dataset generalization without fine-tuning, we apply the PanNuke-trained RayCastED-B directly to MoNuSAC, panopTILs, and PUMA. Table 4.6 reports the zero-shot performance in terms of AJI, bPQ, and F1.

Table 4.6. Zero-shot generalization results. RayCastED-B is trained on PanNuke and evaluated on three unseen datasets without fine-tuning.

Dataset	**AJI**	**bPQ**	**F1**	**P**	**R**
PUMA	0.6336	0.6463	0.7774	0.7579	0.7979
panopTILs	0.2673	0.2554	0.5057	0.3683	0.8063
MoNuSAC	0.3478	0.2648	0.5636	0.5907	0.5390

RayCastED-B generalizes well to PUMA, achieving performance comparable to the in-domain PanNuke results. For panopTILs and MoNuSAC, however, the zero-shot performance indicates that domain-specific retraining is necessary to bridge the distribution gap. In general, fine-tuning or retraining RayCastED on the target dataset is recommended for optimal performance in a new clinical domain.

4.2 Edge Deployment Evaluation

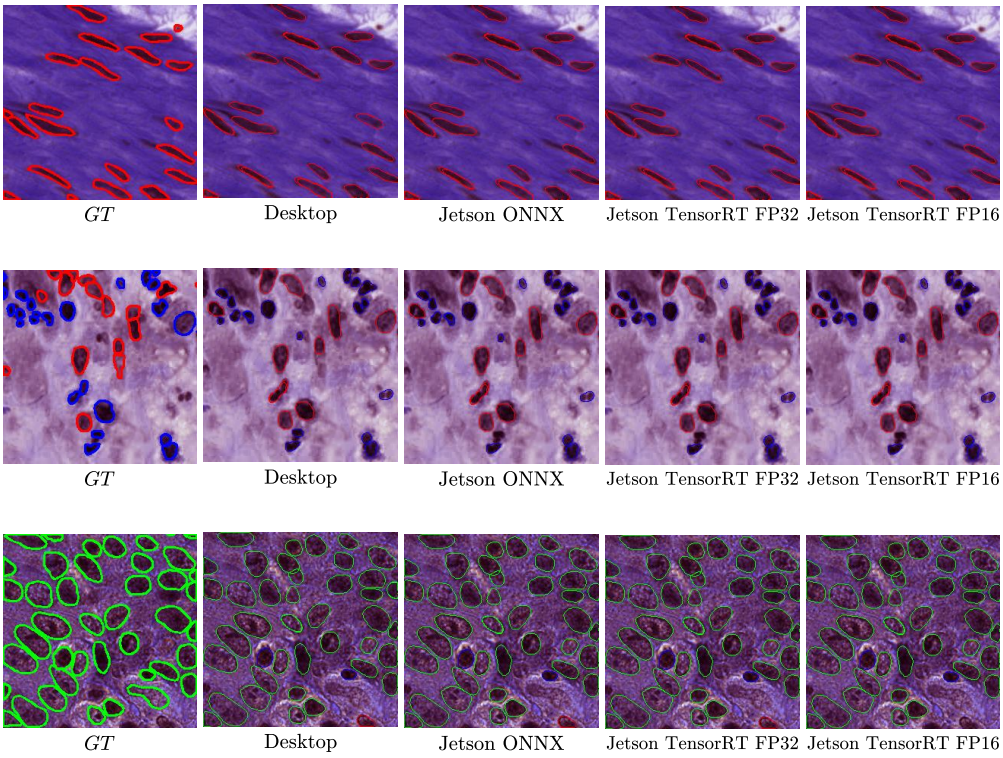


Figure 4.19. Comparison of edge deployment output across platforms.

To assess the practicality of RayCastED for clinical deployment on resource-constrained hardware, we benchmark the smallest variant (RayCastED-S) on two platforms: a desktop GPU (RTX 5070 Ti) and an embedded AI accelerator (Jetson Orin Nano). Table 4.7 compares accuracy and throughput across five configurations.

Table 4.7. Edge deployment evaluation of RayCastED-B across GPU and embedded platforms. All measurements are taken over the PanNuke test set with batch size 1. The best value in each column is shown in **bold**.

Configuration	AJI	bPQ	mPQ	F1	Precision	Recall	Predicted Nuclei	Runtime (ms)
RTX 5070 Ti, PyTorch	0.5715	0.6024	0.4526	0.7409	0.7419	0.7400	**65679 / 65847**	3.59
RTX 5070 Ti, TensorRT	0.5700	0.6003	0.4735	0.7745	0.8067	**0.7448**	60796 / 65847	**2.70**
Jetson Orin Nano, ONNX	0.5706	0.5986	0.4763	0.7748	0.8169	0.7368	–	88.2
Jetson Orin Nano, TF32	**0.5819**	**0.6118**	**0.4897**	0.7749	**0.8170**	0.7369	59394 / 65848	38.1
Jetson Orin Nano, TF16	0.5817	0.6112	0.4894	**0.7753**	0.8140	0.7401	–	29.2
RTX 2080 Super, TensorRT	0.5694	0.5983	0.4746	0.7746	0.8149	0.7381	59640 / 65848	38.8

The PyTorch baseline on the RTX 5070 Ti achieves near-exact predicted nuclei counts, confirming that the end-to-end inference pipeline operates correctly. When com-

piled with TensorRT, the RTX 5070 Ti reaches a per-image latency of 2.70 ms, representing a substantial speedup over PyTorch. Unexpectedly, TensorRT deployment on the Jetson Orin Nano yields consistently higher AJI, bPQ, and mPQ than the desktop GPU across both FP16 and TF32 configurations. This is unlikely to stem from reduced precision, since TF32 and FP16 produce near-identical metrics. A plausible explanation is that TensorRT's aggressive graph-level optimizations for the Jetson target architecture restructure the operator execution order in a way that benefits the model numerically. The absence of such gains on the older RTX 2080 Super (similarly running TensorRT) supports the hypothesis that the effect is architecture-specific rather than a universal property of engine compilation.

4.3 Ablation Study

To isolate the contribution of key design choices, we conduct two controlled experiments using RayCastED-B as the base configuration.

4.3.1 Study on Number of Rays

We vary the number of rays R while keeping all other hyperparameters fixed, measuring both detection accuracy and polygon fidelity. Table 4.8 reports the results for $R \in \{8, 16, 32, 64\}$.

Table 4.8. Ablation study on the number of rays R. All experiments use RayCastED-B with identical hyperparameters except R. Best per column in **bold**.

R	AJI	bPQ	mPQ	F1	Precision	Recall
8	0.5646	0.5902	0.4394	0.7373	0.7334	**0.7412**
16	0.5749	0.6019	**0.4606**	0.7387	0.7393	0.7381
32	**0.5759**	**0.6040**	0.4533	0.7396	**0.7455**	0.7339
64	0.5715	0.6024	0.4526	**0.7409**	0.7419	0.7400
STD	0.0051	0.0063	0.0088	0.0015	0.0051	0.0032

Surprisingly, increasing the number of rays from $R = 8$ to $R = 64$ yields only a modest overall performance improvement. The most notable gain occurs in mPQ, suggesting that the angular resolution of the ray representation primarily affects how the model resolves individual nuclei instances rather than detection accuracy. Beyond $R = 16$, the returns diminish, with $R = 32$ and $R = 64$ producing nearly indistinguishable results. Additionally, the standard deviations across all ray configurations do not differ significantly, indicating that the ray count has minimal influence on prediction consistency. This however contrasts with the representation-level results in Table 4.1, where increasing the ray count from $R = 8$ to $R = 64$ substantially improved mask IoU at the geometric

level. The fact that this geometric improvement does not proportionally translate to model performance reveals a current limitation: the network is not able to exploit the higher angular precision offered by denser ray configurations.

4.3.2 Study on Different Wavelet

We compare the Haar wavelet used in ResoConv against alternative wavelet families and a strided-convolution baseline. Table 4.9 reports detection and segmentation metrics for each configuration.

Table 4.9. Ablation study on the wavelet transform used in ResoConv. All experiments use RayCastED-B with $R = 64$. Best per column in **bold**.

Wavelet	**AJI**	**bPQ**	**mPQ**	**F1**	**Precision**	**Recall**
Haar (db1)	0.5715	**0.6024**	**0.4526**	**0.7409**	**0.7419**	**0.7400**
Daubechies-2 (db2)	**0.5722**	0.6001	0.4435	0.7328	0.7269	0.7388
Daubechies-4 (db4)	0.5651	0.5905	0.4379	0.7148	0.7046	0.7252

The best performance consistently comes from the Haar (db1) wavelet. Increasing the number of filter taps (db2, db4) introduces feature smoothing that attenuates the fine-grained edges, chromatin texture, and boundary transitions that distinguish individual nuclei. This smoothing removes discriminative high-frequency content essential for accurate detection and segmentation, resulting in regressed performance on all metrics.

CHAPTER V
CONCLUSIONS AND FUTURE RESEARCH

5.1 Conclusions

This thesis presented RayCastED, a lightweight, fully convolutional, end-to-end model for nuclei detection and segmentation in histopathology images. Unlike prior methods that rely on dense per-pixel prediction and handcrafted post-processing (NMS, watershed, or flow-field), RayCastED formulates the task as a sparse regression problem on an anchor grid and produces final instance masks in a single forward pass. The key findings are summarized below.

Detection and Segmentation Accuracy.

RayCastED achieves competitive performance against state-of-the-art methods on the PanNuke benchmark. The largest variant (RayCastED-L, 9.44M parameters) achieves metrics within 10% of the best result in every accuracy column (AJI, bPQ, mPQ, F1, Precision, Recall) while requiring a fraction of the parameter count and GFLOPs of the most accurate competitor (LKCell, 163.8M parameters). The post-processing-free design enables linear time complexity, with inference speed that scales with the number of anchor positions rather than the number of detected objects. The ablation studies confirm that the Haar wavelet DWT (ResoConv) and the split classification head are the most impactful architectural components, while the ray count ablation indicates that the model does not yet fully exploit higher angular precision – a current limitation that motivates further research.

Edge Deployment Feasibility.

The smallest variant (RayCastED-S, 0.81M parameters) demonstrates viable inference on the Jetson Orin Nano embedded AI accelerator. The NMS-free design ensures that inference complexity is image-size agnostic, determined solely by the fixed anchor grid at a given input resolution. This property enables throughput-consistent deployment across varying tile dimensions, a practical requirement for clinical whole-slide image analysis pipelines. TensorRT-optimized inference achieves real-time or near-real-time throughput on both desktop and embedded platforms.

Generalization.

Zero-shot evaluation on PUMA, panopTILs, and MoNuSAC reveals reasonable generalization of the PanNuke-trained model to PUMA (melanoma TILs), but a substantial

performance drop on panopTILs and MoNuSAC, indicating that tissue-level and staining-protocol domain gaps require fine-tuning or retraining for target clinical domains. The RayCast representation itself is not dataset-specific and can be generated from any polygon annotation format, making the overall pipeline applicable to diverse data sources.

5.2 Future Research

Several directions warrant further investigation:

1. **Adaptive ray density.** The current fixed ray count ($R = 64$) applies uniformly to all nuclei regardless of size. An adaptive scheme that assigns more rays to larger nuclei and fewer to smaller cells could reduce computational cost while maintaining boundary accuracy for large, clinically relevant nuclei.
2. **Non-star-convex representations.** The star-convex constraint limits the RayCast representation to simple polygon shapes. Extending the representation to handle concavities and irregular boundaries would broaden applicability to complex nuclear morphologies.
3. **Domain adaptation and fine-tuning.** The generalization results indicate that domain-specific retraining is necessary for robust performance on new tissue types and staining protocols. Lightweight fine-tuning strategies, such as adapter layers or prompt-based adaptation, could enable rapid deployment to new clinical settings while preserving the backbone's learned representations.
4. **Investigating the Jetson TensorRT accuracy anomaly.** The Jetson Orin Nano TensorRT configurations consistently outperform the desktop RTX 5070 Ti on accuracy metrics. Since quantization alone does not account for this discrepancy, a systematic investigation of architecture-specific TensorRT graph optimizations, memory layout differences, and numerical precision paths could reveal optimizations generalizable to other deployments.
5. **INT8 quantization.** Further acceleration via INT8 quantization could reduce inference latency and memory footprint on edge devices. Quantization-aware training or calibration-based post-training quantization should be evaluated for impact on detection and segmentation accuracy.
6. **Whole-slide image integration.** The tile-based processing pipeline can be extended to end-to-end WSI analysis by coupling RayCastED with a tissue detection front-end and a tile-level result stitching and deduplication back-end. This would enable direct clinical deployment for tasks such as TILs scoring and tumor microenvironment analysis.
7. **Extension to other medical tasks.** The RayCast representation and the post-

processing-free detection pipeline are not specific to nuclei. The same framework can be applied to other medical instance-level tasks such as glomeruli detection in renal pathology, gland segmentation in colon histology, and cell counting in cytology.

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LAMPIRAN

L.1 Isi Lampiran

Lampiran bersifat opsional bergantung hasil kesepakatan dengan pembimbing dapat berupa:

1. Bukti pelaksanaan Kuesioner seperti pertanyaan kuesioner, resume jawaban responden, dan dokumentasi kuesioner.
2. Spesifikasi Aplikasi atau Sistem yang dikembangkan meliputi spesifikasi teknis aplikasi, tautan unduh aplikasi, manual penggunaan aplikasi, hingga screenshot aplikasi.
3. Cuplikan kode yang sekiranya penting dan ditambahkan.
4. Tabel yang terlalu panjang yang masih diperlukan tetapi tidak memungkinkan untuk ditayangkan di bagian utama skripsi.
5. Gambar-gambar pendukung yang tidak terlalu penting untuk ditampilkan di bagian utama. Akan tetapi, mendukung argumentasi/pengamatan/analisis.
6. Penurunan rumus-rumus atau pembuktian suatu teorema yang terlalu panjang dan terlalu teknis sehingga Anda berasumsi bahwa pembaca biasa tidak akan menelaah lebih lanjut. Hal ini digunakan untuk memberikan kesempatan bagi pembaca tingkat lanjut untuk melihat proses penurunan rumus-rumus ini.

LAMPIRAN

L.2 Panduan Latex

L.2.1 Syntax Dasar

L.2.1.1 Penggunaan Sitasi

Contoh penggunaan sitasi [41, 42] [43] [44] [45] [46, 47]

L.2.1.2 Penulisan Gambar

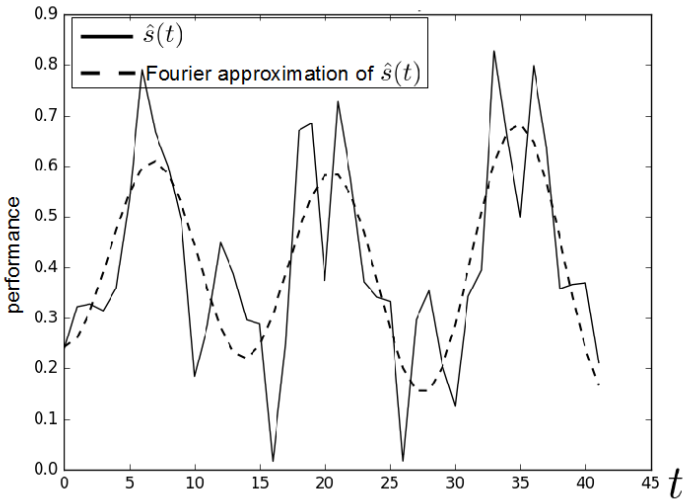

Figure L.1. Contoh gambar.

Contoh gambar terlihat pada Gambar L.1. Gambar diambil dari [47].

L.2.1.3 Penulisan Tabel

Table L.1. Tabel ini

ID	Tinggi Badan (cm)	Berat Badan (kg)
A23	173	62
A25	185	78
A10	162	70

Contoh penulisan tabel bisa dilihat pada Tabel L.1.

L.2.1.4 Penulisan formula

Contoh penulisan formula

$$L_{\psi_z} = \{t_i \mid v_z(t_i) \leq \psi_z\} \tag{1}$$

Contoh penulisan secara *inline*: $PV = nRT$. Untuk kasus-kasus tertentu, kita membutuhkan perintah "mathit" dalam penulisan formula untuk menghindari adanya jeda saat penulisan formula.

Contoh formula **tanpa** menggunakan "mathit": $PVA = RTD$

Contoh formula **dengan** menggunakan "mathit": $PVA = RTD$

L.2.1.5 Contoh list

Berikut contoh penggunaan list

1. First item
2. Second item
3. Third item

L.2.2 Blok Beda Halaman

L.2.2.1 Membuat algoritma terpisah

Untuk membuat algoritma terpisah seperti pada contoh berikut, kita dapat memanfaatkan perintah *algstore* dan *algrestore* yang terdapat pada paket *algcompatible*. Pada dasarnya, kita membuat dua blok algoritma dimana blok pertama kita simpan menggunakan *algstore* dan kemudian di-restore menggunakan *algrestore* pada algoritma kedua. Perintah tersebut dimaksudkan agar terdapat kesinamungan antara kedua blok yang sejatinya adalah satu blok.

Algorithm 1 Contoh algorima

```
1: procedure CREATESET(v)
2:     Create new set containing v
3: end procedure
```

Pada blok algoritma kedua, tidak perlu ditambahkan caption dan label, karena sudah menjadi satu bagian dalam blok pertama. Pembagian algoritma menjadi dua bagian ini berguna jika kita ingin menjelaskan bagian-bagian dari sebuah algoritma, maupun untuk memisah algoritma panjang dalam beberapa halaman.

```
4: procedure CONCATSET(v)
5:     Create new set containing v
6: end procedure
```

L.2.2.2 Membuat tabel terpisah

Untuk membuat tabel panjang yang melebihi satu halaman, kita dapat mengganti kombinasi *table* + *tabular* menjadi *longtable* dengan contoh sebagai berikut.

Table L.2. Contoh tabel panjang

header 1	header 2
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar
foo	bar

L.2.2.3 Menulis formula terpisah halaman

Terkadang kita butuh untuk menuliskan rangkaian formula dalam jumlah besar sehingga melewati batas satu halaman. Solusi yang digunakan bisa saja dengan memindahkan satu blok formula tersebut pada halaman yang baru atau memisah rangkaian formula menjadi dua bagian untuk masing-masing halaman. Cara yang pertama mungkin akan menghasilkan alur yang berbeda karena ruang kosong pada halaman pertama akan diisi oleh teks selanjutnya. Sehingga di sini kita dapat memanfaatkan *align* yang sudah diatur dengan mode *allowdisplaybreaks*. Penggunakan *align* ini memungkinkan satu rangkaian formula terpisah berbeda halaman.

Contoh sederhana dapat digambarkan sebagai berikut.

$$\begin{aligned} x &= y^2 \\ x &= y^3 \\ a + b &= c \\ x &= y - 2 \\ a + b &= d + e \\ x^2 + 3 &= y \\ a(x) &= 2x \\ b_i &= 5x \end{aligned} \tag{2}$$

$$10x^2 = 9x$$
$$2x^2 + 3x + 2 = 0$$
$$5x - 2 = 0$$
$$d = \log x$$
$$y = \sin x$$

LAMPIRAN

L.3 Format Penulisan Referensi

Penulisan referensi mengikuti aturan standar yang sudah ditentukan. Untuk internasionalisasi DTETI, maka penulisan referensi akan mengikuti standar yang ditetapkan oleh IEEE (*International Electronics and Electrical Engineers*). Aturan penulisan ini bisa diunduh di http://www.ieee.org/documents/ieeecitationref.pdf. Gunakan Mendeley sebagai *reference manager* dan *export* data ke format Bibtex untuk digunakan di Latex.

Berikut ini adalah sampel penulisan dalam format IEEE:

L.3.1 Book

Basic Format:

[1] J. K. Author, "Title of chapter in the book," in Title of His Published Book, xth ed. City of Publisher, Country: Abbrev. of Publisher, year, ch. x, sec. x, pp. xxx–xxx.

Examples:

[1] B. Klaus and P. Horn, Robot Vision. Cambridge, MA: MIT Press, 1986.

[2] L. Stein, "Random patterns," in Computers and You, J. S. Brake, Ed. New York: Wiley, 1994, pp. 55-70.

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Polymers of Hexadromicon, J. Peters, Ed., 2nd ed. New York: McGraw-Hill, 1964, pp. 15-64.

L.3.2 Handbook

Basic Format:

[1] Name of Manual/Handbook, x ed., Abbrev. Name of Co., City of Co., Abbrev. State, year, pp. xx-xx.

Examples:

[1] Transmission Systems for Communications, 3rd ed., Western Electric Co., Winston Salem, NC, 1985, pp. 44-60.

[2] Motorola Semiconductor Data Manual, Motorola Semiconductor Products Inc., Phoenix, AZ, 1989.

[3] RCA Receiving Tube Manual, Radio Corp. of America, Electronic Components and Devices, Harrison, NJ, Tech. Ser. RC-23, 1992.

Conference/Prosiding

Basic Format:

[1] J. K. Author, "Title of paper," in Unabbreviated Name of Conf., City of Conf., Abbrev. State (if given), year, pp.xxx-xxx.

Examples:

[1] J. K. Author [two authors: J. K. Author and A. N. Writer] [three or more authors: J. K. Author et al.], "Title of Article," in [Title of Conf. Record as], [copyright year] © [IEEE or applicable copyright holder of the Conference Record]. doi: [DOI number]

Sumber Online/Internet

Basic Format:

[1] J. K. Author. (year, month day). Title (edition) [Type of medium]. Available: http://www.(URL)

Examples:

[1] J. Jones. (1991, May 10). Networks (2nd ed.) [Online]. Available: http://www.atm.com

Skripsi, Tesis dan Disertasi

Basic Format:

[1] J. K. Author, "Title of thesis," M.S. thesis, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.

[2] J. K. Author, “Title of dissertation,” Ph.D. dissertation, Abbrev. Dept., Abbrev. Univ., City of Univ., Abbrev. State, year.

Examples:

[1] J. O. Williams, “Narrow-band analyzer,” Ph.D. dissertation, Dept. Elect. Eng., Harvard Univ., Cambridge, MA, 1993. [2] N. Kawasaki, “Parametric study of thermal and chemical nonequilibrium nozzle flow,” M.S. thesis, Dept. Electron. Eng., Osaka Univ., Osaka, Japan, 1993

L.4 Contoh Source Code

L.4.1 Sample algorithm

Algorithm 2 Kruskal's Algorithm

```
1:  procedure MAKESET(v)
2:      Create new set containing v
3:  end procedure
4:
5:  function FINDSET(v)
6:      return a set containing v
7:  end function
8:
9:  procedure UNION(u,v)
10:     Unites the set that contain u and v into a new set
11: end procedure
12:
13: function KRUSKAL(V, E, w)
14:     A ← {}
15:     for each vertex v in V do
16:         MakeSet(v)
17:     end for
18:     Arrange E in increasing costs, ordered by w
19:     for each (u,v) taken from the sorted list do
20:         if FindSet(u) ≠ FindSet(v) then
21:             A ← A ∪ {(u, v)}
22:             Union(u, v)
23:         end if
24:     end for
25:     return A
26: end function
```

L.4.2 Sample Python code

```
import numpy as np

def incmatrix(genl1,genl2):
  m = len(genl1)
  n = len(genl2)
  M = None #to become the incidence matrix
  VT = np.zeros((n*m,1), int)  #dummy variable

  #compute the bitwise xor matrix
  M1 = bitxormatrix(genl1)
  M2 = np.triu(bitxormatrix(genl2),1)

  for i in range(m-1):
    for j in range(i+1, m):
      [r,c] = np.where(M2 == M1[i,j])
      for k in range(len(r)):
        VT[(i)*n + r[k]] = 1;
        VT[(i)*n + c[k]] = 1;
        VT[(j)*n + r[k]] = 1;
        VT[(j)*n + c[k]] = 1;

  if M is None:
    M = np.copy(VT)
  else:
    M = np.concatenate((M, VT), 1)

  VT = np.zeros((n*m,1), int)

  return M
```

L.4.3 Sample Matlab code

```
function X = BitXorMatrix(A,B)
%function to compute the sum without charge of two vectors

  %convert elements into usigned integers
  A = uint8(A);
  B = uint8(B);

  m1 = length(A);
  m2 = length(B);
  X = uint8(zeros(m1, m2));
  for n1=1:m1
    for n2=1:m2
      X(n1, n2) = bitxor(A(n1), B(n2));
    end
  end
```